Data sources

For this exercise two data sources have been chosen. Both of them are Wikipedia categories, which are Twitter and YouTube. The corresponding links are following:

- https://en.wikipedia.org/wiki/Category:Twitter
- https://en.wikipedia.org/wiki/Category:YouTube

Setup

This part discusses in short the setup in order to be able to run the following script

The following code install the required dependencies.

In [1]:
```
!pip install matplotlib
!pip install numpy
!pip install pandas
!pip install wikipedia-api
```

```
Requirement already satisfied: matplotlib in ./venv/lib/python3.10/site-
packages (3.6.2)
Requirement already satisfied: pyparsing>=2.2.1 in ./venv/lib/python3.10
/site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: cycler>=0.10 in ./venv/lib/python3.10/sit
e-packages (from matplotlib) (0.11.0)
Requirement already satisfied: kiwisolver>=1.0.1 in ./venv/lib/python3.1
0/site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: numpy>=1.19 in ./venv/lib/python3.10/site
-packages (from matplotlib) (1.23.5)
Requirement already satisfied: python-dateutil>=2.7 in ./venv/lib/python
3.10/site-packages (from matplotlib) (2.8.2)
Requirement already satisfied: pillow>=6.2.0 in ./venv/lib/python3.10/si
te-packages (from matplotlib) (9.3.0)
Requirement already satisfied: fonttools>=4.22.0 in ./venv/lib/python3.1
0/site-packages (from matplotlib) (4.38.0)
Requirement already satisfied: contourpy>=1.0.1 in ./venv/lib/python3.10
/site-packages (from matplotlib) (1.0.6)
Requirement already satisfied: packaging>=20.0 in ./venv/lib/python3.10/
site-packages (from matplotlib) (21.3)
Requirement already satisfied: six>=1.5 in ./venv/lib/python3.10/site-pa
ckages (from python-dateutil>=2.7->matplotlib) (1.12.0)
WARNING: You are using pip version 21.2.3; however, version 22.3.1 is av
ailable.
You should consider upgrading via the '/Users/gcm/PyProjects/CLT/venv/bi
n/python3.10 -m pip install --upgrade pip' command.
Requirement already satisfied: numpy in ./venv/lib/python3.10/site-packa
```

```
ges (1.23.5)
WARNING: You are using pip version 21.2.3; however, version 22.3.1 is av
ailable.
You should consider upgrading via the '/Users/gcm/PyProjects/CLT/venv/bi
n/python3.10 -m pip install --upgrade pip' command.
Requirement already satisfied: pandas in ./venv/lib/python3.10/site-pack
ages (1.5.2)
Requirement already satisfied: pytz>=2020.1 in ./venv/lib/python3.10/sit
e-packages (from pandas) (2022.6)
Requirement already satisfied: numpy>=1.21.0 in ./venv/lib/python3.10/si
te-packages (from pandas) (1.23.5)
Requirement already satisfied: python-dateutil>=2.8.1 in ./venv/lib/pyth
on3.10/site-packages (from pandas) (2.8.2)
Requirement already satisfied: six>=1.5 in ./venv/lib/python3.10/site-pa
ckages (from python-dateutil>=2.8.1->pandas) (1.12.0)
WARNING: You are using pip version 21.2.3; however, version 22.3.1 is av
ailable.
You should consider upgrading via the '/Users/gcm/PyProjects/CLT/venv/bi
n/python3.10 -m pip install --upgrade pip' command.
Requirement already satisfied: wikipedia-api in ./venv/lib/python3.10/si
te-packages (0.5.4)
Requirement already satisfied: requests in ./venv/lib/python3.10/site-pa
ckages (from wikipedia-api) (2.28.1)
Requirement already satisfied: idna<4,>=2.5 in ./venv/lib/python3.10/sit
e-packages (from requests->wikipedia-api) (3.4)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in ./venv/lib/pytho
n3.10/site-packages (from requests->wikipedia-api) (1.24.3)
Requirement already satisfied: charset-normalizer<3,>=2 in ./venv/lib/py
thon3.10/site-packages (from requests->wikipedia-api) (2.1.1)
Requirement already satisfied: certifi>=2017.4.17 in ./venv/lib/python3.
10/site-packages (from requests->wikipedia-api) (2022.9.24)
WARNING: You are using pip version 21.2.3; however, version 22.3.1 is av
ailable.
You should consider upgrading via the '/Users/gcm/PyProjects/CLT/venv/bi
n/python3.10 -m pip install --upgrade pip' command.
```

The next code block imports are required packages which have been installed prior.

In [91]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from PIL import Image
import seaborn as sns
from sklearn.feature_extraction.text import CountVectorizer
from textblob import TextBlob
import wikipediaapi
from wordcloud import WordCloud

%matplotlib inline
plt.rcParams['figure.figsize'] = [12, 8]
plt.rcParams['figure.dpi'] = 100

pd.options.display.max_colwidth = 100
pd.options.display.max_columns = 30

pd.set_option('display.max_columns', None)
pd.set_option('max_colwidth', None)
```

Retrieving Data from Wikipedia API

Accessing the Wikipedia API and saving the pages 'Category:Twitter' and 'Category:YouTube' to the variables 'cat_twitter' and 'cat_youtube'. This variable will be used in the next step.

In [3]:
```
wiki = wikipediaapi.Wikipedia('en')
cat_twitter = wiki.page('Category:Twitter')
twitter = wiki.page('Twitter')
cat_youtube = wiki.page('Category:YouTube')
youtube = wiki.page('YouTube')
```

In [4]:
```
cat_twitter.text
```

Out[4]:
```
'Twitter and related articles, as well as Twitter powered blogs.'
```

In [5]:
```
cat_youtube.text
```

Out[5]:
```
'This category covers articles on the subject of YouTube.'
```

The function below creates a dictionary. The input are articles which are listet in the Twitter/YouTube category. The name of the article is the key. As value is passed a list, which contains the summary of the article and the amount of links which are present on each article. The data structure looks like this: `{key1: [value1.1, value1.2], key2: [value2.1, value.2], ...}`

In [6]:
```
def categories(name):
        mdict = {}
        for c in name.values():
                l = []
                if not 'Category:' in c.title:
                        link_count = len(wiki.page(c.title).links)
                        l.append(c.summary)
                        l.append(link_count)
                        mdict[c.title] = l
        return mdict
```

Twitter

The wikipedia page is passed to the function. Note, in this step the method 'categorymembers' is used to access all members of the corresponding category

In [7]:
```
twitter_dict = categories(cat_twitter.categorymembers)
```

The keys of the dict are printed out to gain an overview of the article names which have been retrieved

In [8]:
```
twitter_dict.keys()
```

Out[8]:
```
dict_keys(['Twitter', 'Aufschrei', '100 Nichi Go ni Shinu Wani', 'Artifi
cial Intelligence for Digital Response', 'Justin Bieber', 'Billboard Twi
tter Real-Time', 'Black Twitter', 'Bluesky (protocol)', 'Bootstrap (fron
t-end framework)', 'Breaking Tweets', 'Chaos with Ed Miliband', 'Dogs at
polling stations', 'FlockDB', 'Frogtwitter', 'Hashtag', 'History of Twit
ter', 'Internet activism during the 2009 Iranian election protests', '20
09 Iranian presidential election protests', 'ISIS-chan', 'KHive', 'List
of Twitter features', 'The Magical Moment', 'Mention (blogging)', 'Mikey
y', 'Museum Week', 'Republik Twitter', 'Roger Cosmonkey', 'Rotation Cura
tion', 'Safe Space (South Park)', 'Takahiro Shiraishi', 'Shorty Awards',
'Skank Hunt', 'Snowflake ID', 'Elon Musk', 'Stan Twitter', 'Such Tweet S
orrow', 'Tawawa on Monday', 'Twitter trends', 'Twitter bomb', 'Twitter d
iplomacy', 'Twitter Power', 'Twitter Revolution', 'Twitter usage', 'Twit
ter verification', 'Twitter Zero', 'Twitterature', 'Unknown Number', 'Wh
yIStayed/WhyILeft'])
```

The dict can be accessed by the following code. The key has to be passed, additionally the index (in this case 0) to access the summary of the article. The meaning of '[:100]' is to print only the first 100 characters of the value.

In [9]:
```
twitter_dict['Twitter'][0][:100]
```

Out[9]:
```
'Twitter is a microblogging, social networking service owned by American
company Twitter, Inc., on wh'
```

In order to see how many links are on the Twitter article the dict can be accessed with index 1.

In [10]:
```
twitter_dict['Twitter'][1]
```

Out[10]:
```
953
```

YouTube

The same procedure is performed with the second topic, which is YouTube

In [11]:
```
youtube_dict = categories(cat_youtube.categorymembers)
```

In [12]:
```
youtube_dict.keys()
```

Out[12]:
```
dict_keys(['YouTube', 'List of influencer boxing matches', '2013 YouTube
Music Awards', '2015 YouTube Music Awards', 'Alternative Influence: Broa
dcasting the Reactionary Right on YouTube', 'Bob Thunder: Internet Assas
sin', 'BookTube', 'Censorship of YouTube', 'Steve Chen', 'Collab house',
'Content house', 'Content ID (system)', 'YouTube Creator Awards', 'Creat
or Clash', 'Deen The Great vs Walid Sharks', 'Face to Face with "Weird A
l" Yankovic', 'FameBit', 'Google Preferred', 'Chad Hurley', 'InfoChammel
', 'Invidious', 'Jay Swingler vs Cherdleys', 'Justin Bieber: Seasons', '
Salar Kamangar', 'Jawed Karim', 'Keeping It Real Art Critics', 'Kevin Ha
rt: What The Fit', 'YouTube Kids', 'KSI vs Dillon Danis', 'KSI vs Logan
Paul', 'KSI vs Swarmz & Luis Alcaraz Pineda', 'KSI vs. Logan Paul II', "
Let's Play", 'Lion Versus', 'Mothership (composition)', 'Multi-channel n
etwork', 'YouTube Music Awards', 'Alex Okosi', 'Petit Tube', 'PewDiePie
vs T-Series', 'Playlist Live', 'Project for Awesome', 'Remastering Proje
ct', 'Reply girl', 'The Slot (TV series)', 'Social impact of YouTube', '
SponsorBlock', 'Tubecon', 'Vevo', 'VidIQ', 'Why Pay More?', 'Susan Wojci
cki', 'Yattee', 'You (Time Person of the Year)', 'The You Generation', '
YouTomb', 'YouTube (channel)', 'YouTube and privacy', 'YouTube Comedy We
ek', 'Comparison of YouTube downloaders', 'YouTube in education', 'YouTu
be Instant', 'YouTube Live', 'YouTube Music', 'YouTube Next Lab and Audi
ence Development Group', 'YouTube Original Channel Initiative', 'YouTube
poop', 'Template:YouTube Premium films and documentaries', 'Template:You
Tube Premium original programming', 'YouTube Shorts', 'YouTube Space', '
YouTube Space Lab', 'YouTube Symphony Orchestra', 'YouTube Theater', 'Yo
uTube TV', 'YouTube Vanced', 'Youtube-dl', 'Youtubers Life', 'YouTubers
vs. TikTokers', 'Template:YTMAs'])
```

In [13]:
```
youtube_dict['YouTube'][0][:100]
```

Out[13]:
```
'YouTube is an American online video sharing and social media platform h
eadquartered in San Bruno, Ca'
```

In [14]:
```
youtube_dict['YouTube'][1]
```

Out[14]:
```
1745
```

Saving dict to pandas dataframe

The loop below saves the dict into a dataframe, which is printed afterwards. Additionally, a label for each topic is inserted. This step is executed for Twitter and YouTube.

In [66]:
```
d = []
for i in range(len(twitter_dict)):
        d.append({
                'title': list(twitter_dict.keys())[i],
                'description': list(twitter_dict.values())[i][0],
                'links': list(twitter_dict.values())[i][1],
        })
df_twitter = pd.DataFrame(d)
df_twitter['label'] = 'twitter'
df_twitter
```

Out[66]:

	title	
0	Twitter	Twitter is a microblogging, social networking service owned by Ameri post, like, and retweet tweets, while unregistered users or programmatically via its APIs. Before April 2020, services were a languages in November 2017. Audio and video tweets remain limited t March 2006 and launched in July of that year. Twitter, Inc. is based in S 340 million tweets a day, and the service handled an average of 1.6 billi of the Internet". By the start of 2019, Twitter had more than 330 mi accounts were estimated to make up 15% of Twitter accounts, appro Musk, making it one of the biggest deals to take a company private. Twitter has been cr
1	Aufschrei	#Aufschrei ("outcry") is a German hashtag which went viral on the s Tweets began to appear in response to the publication of an article in w Democratic Party (FDP); within
2	100 Nichi Go ni Shinu Wani	100 Nichi Go ni Shinu Wani (100日後に死ぬワニ, "A Crocodile Who Will a crocodile's last 100 days before his death, was self-published on Twi 8, 2020. An anime film adaptation produced by TIA, titled 100 Nichik
3	Artificial Intelligence for Digital Response	Artificial Intelligence for Digital Response (AIDR) is a free and open sou It has been developed by the Qatar Computing Research Institute and his team "have developed novel computational techniques and technolog - according to him the system "combines human intelligence
		Justin Drew Bieber (BEE-bər; born March 1, 1994) is a Canadian sing music. He was discovered by American record executive Scooter Braur (2009) and soon establishing himself as a teen idol.\nBieber achieved Billboard 200, making him the youngest solo male act to top the certified singles of all time in the US. His second studio album, Un experimented with dance-pop in his third studio album, Believe (2012), Bieber was involved in multiple controversies and legal issues, which h won the Grammy Award for Best Dance/Electronic Recording. Th number-one singles: "Love Yourself", "Sorry", and "What Do You Mea numerous collaborations between 2016 and 2017, including "I'm the Or

4	Justin Bieber	the first artist in history to chart new number-one songs in consecut Latin Grammy Award. In 2019, he released a country collaboration with D released his R&B-led fifth studio album, Changes, which debuted at nu US Billboard Hot 100. Bieber returned to his pop roots on his sixth st him the youngest soloist to have eight US number-one albums, a record his eighth US number-one single.Bieber is one of the best-selling m certifications from the RIAA for "Baby", "Sorry" and "Despacito". He Brit Awards, one Bambi Award, 26 Billboard Music Awards, 18 America for a male individual), and 33 Guinness World Records. Time named
5	Billboard Twitter Real-Time	The Billboard Twitter Real-Time charts were four interactive Billboard mu based on how often they are mentioned in messages called "tweets" se Tracks" and "Weekly Emerging Artists" were reported to define how far U.S. The first two charts updated on a "real-time" minute-by-minu "Trending 140" and real-time charts were discontinued and the Emerg rising acts of each week.In October 2021, Billboard and Twitter launch first ch
6	Black Twitter	Black Twitter is an internet community largely consisting of African-Am Feminista Jones described it in Salon as "a collective of active, primari
7	Bluesky (protocol)	Bluesky is an initiative to develop a decentralized social network prot Bluesky
8	Bootstrap (front-end framework)	Bootstrap is a free and open-source CSS framework directed at re templates for typography, forms, buttons, navigation, and other inter
9	Breaking Tweets	Breaking Tweets was an international journalism news Web site that tra "hyperlocal gone global," personalizing major news events throug Breaking Tweets Sports, Breaking Tweets Entertainment, and Br
10	Chaos with Ed Miliband	"Chaos with Ed Miliband" is a phrase from a 2015 tweet by David Cam before the 2015 United Kingdom general election – Cameron, the l Government" with him, or "chaos with Ed Miliband", who was leadin
11	Dogs at polling stations	Dogs at polling stations or #dogsatpollingstations is a popular hashtag dogs are photographed waiting for their owners outside the polling s general elections of 2015 and 2017.The 2019 United Kingdom gener wearing a Santa hat. Politicians who participated included Boris Johns Carrot, as they do not have a dog. Other animals, such as horses, also
12	FlockDB	FlockDB is an open-source distributed, fault-tolerant graph database e.g. followings and favorites. FlockDB differs from other graph database primary use-case for Redis sets. Since it is still in the process of bein FlockDB was posted on GitHub shortly after Twitter released its Gizza
13	Frogtwitter	#Frogtwitter is a network of loosely-connected pseudonymous Twit could also deri
14	Hashtag	A hashtag is a metadata tag that is prefaced by the hash (also know such as Twitter or Instagram as a form of user-generated tagging #bluesky returns all posts that have been tagged with that term proposed by American blogger and product consultant Chris Messina in owned by no one". Hashtags became entrenched in the culture of T English Dictionary as "a word or phrase with the symbol # in

15	History of Twitter	
16	Internet activism during the 2009 Iranian election protests	Internet activism and, specifically, social networking has been instrume video, and Twitter, Facebook, and blogs have been places for prot organi
17	2009 Iranian presidential election protests	After incumbent president Mahmoud Ahmadinejad declared v candidates Mir-Hossein Mousavi and Mehdi Karroubi. The pro proponents, reflecting Mousavi's campaign theme, and Persian Awak that incumbent President Mahmoud Ahmadinejad won nearly 63 percent manipulated and the election was rigged, with Rezaee and Mousavi appeal against the result to the Guardian Council on 14 June. Ayato Movement leaders. Ahmadinejad called the election "completely free" a match".Due to the sporadic cases of violence present at the pro sprayed, arrested and tortured, and even shot in some cases. The mos around the world. Opposition groups also reported thousands more w children by the Islamic Revolutionary Guards in prisons such as K reports by Mousavi supporters allege at least 72 deaths (over t documents citing death by heart attack or meningitis. Iranian authorities Green Movement developed during these protests. It was
18	ISIS-chan	ISIS-chan (ISIS（アイシス）ちゃん, aishisuchan) is a fictional man incident
19	KHive	#KHive is the hashtag used by an informal online community supporting and occasionally transcribed as K-Hive, and refers to the wider online 2020 presidential campaign as an effort to defend Harris from racist
20	List of Twitter features	Twitter is an American microblogging and social networking service on v
21	The Magical Moment	The Magical Moment is a TweetBook by Brazilian author Paulo Coelho.
22	Mention (blogging)	A mention (also known as @replies or tagging) is a means by which a attention to) another user of a social networking or blogging serv
23	Mikeyy	Mikeyy is the name of a computer worm that spread approximately discrete attacks "between 2 AM Saturday April 11, 2009 Pacific tim called StalkDaily.The worm was written by 17-year-old Michael Moon
24	Museum Week	Museum Week is a global event that takes place during one week each collaboration of twelve French museums wishing to expand their audie
25	Republik Twitter	Republik Twitter (stylised as #republiktwitter; meaning The Twitter Rep Aryasatya, it follows a university student in his quest to be with a beau script. Based on one of his earlier blog posts, pre-production for the
26	Roger Cosmonkey	Roger Cosmonkey is a series of webcomics created by filmmaker Cory E Cosmonkey 2: Mega-Pigeon.\nBoth series were posted on Edwards' included a link to a single comic strip in the series.\nIt has been describ feature film of the television series Fraggle Rock expired. He fou
27	Rotation Curation	Rotation Curation, also #RotationCuration, is the concept of rotating the a group,
28	Safe Space (South	"Safe Space" is the fifth episode of the nineteenth season and the 262n

	Park)	The episode premiered on Comedy Central on October
29	Takahiro Shiraishi	Takahiro Shiraishi (白石隆浩, Shiraishi Takahiro, born 9 October 1990) reports at the time of his sentencing. In Zama, Japan,
30	Shorty Awards	The Shorty Awards, also known as the "Shortys", is an annual awards YouTube, Instagram, TikTok, Twitch and the rest of the social web.\nThe recognized content creation on other social networking sites, including live. Each award recognizes the entire body of work of the content cr commented: "The Shortys recognize that social media is about more th
31	Skank Hunt	"Skank Hunt" is the second episode in the twentieth season of the Ame
32	Snowflake ID	Snowflake IDs, or snowflakes, are a form of unique identifier usec a
33	Elon Musk	Elon Reeve Musk (EE-lon; born June 28, 1971) is a business magnate ar Inc.; founder of The Boring Company; co-founder of Neuralink and C $181 billion as of November 18, 2022, Musk is the wealthiest person in th$ $. Musk was born and grew up in Pretoria, South Africa. He attended th$ $- \text{ born mother. Two years later, he matriculated at Queen's Universit}$ $\text{where he received bachelor's degrees in economics and physics. He mov}$ $\text{but dropped out after two days to instead pursue a business career, co} -$ 307 million in 1999. The same year, Musk co-f $1.5 billion. \backslash \text{nIn} 2002, \text{Musk founded SpaceX, an aerospace manu fact}$ $\text{he was an early investor in the electric vehicle manu facturer Tesla Mo}$ $\text{eventually assuming the position of CEO in 2008. In 2006, he helped cre}$ $- \text{ founded OpenAI, a nonprofit artificial intelligence research comp}$ $- \text{computer interfaces — and The Boring Company, a tunnel construc}$ 44 billion. He has proposed a hyperloop high-speed vactrain transpor regarding politics and technology, particularly on Twitter, and is a highl COVID-19 misinformation and promoting a conspiracy theory concer for falsely tweeting that he had secured funding for a private takeover the SEC. Following his failed submarine proposal during the Th
34	Stan Twitter	Stan Twitter is a community of Twitter users that post opinions relate terminology but also for incidents of harass
35	Such Tweet Sorrow	Such Tweet Sorrow was a modern-day adaptation of Shakespear performed the play on Twitter and other web devices. The actors im comments via
36	Tawawa on Monday	Tawawa on Monday (Japanese: 月曜日のたわわ, Hepburn: Getsuyōb Himura has posted an illustration on his Twitter account every December 2016. A second season by Yokohama Animation Labor
37	Twitter trends	On Twitter, a word, phrase, or topic that is mentioned at a greater concerted effort by users or because of an event that prompts people and what people's opinions are about it.Trending topics are sometim
38	Twitter bomb	The term twitter bomb or tweet bomb (also spelled as one word) refers @messages, from multiple accounts, with the goal of advertising a certa
39	Twitter diplomacy	Twitter diplomacy, also called "Twiplomacy" or "hashtag diplomacy," is t their diplomats to conduct diplomatic outreach and public dip

40	Twitter Power	Twitter Power: How to Dominate Your Market One Tweet at a Time Twitter account, market their brands, and gain Twitter followers.\nPu giving r
41	Twitter Revolution	Twitter Revolution is a term used to refer to different revolutions and pr
42	Twitter usage	Since the
43	Twitter verification	Twitter verification is a system intended to communicate the auth distinguish genuine notable account holders, such as celebrities and or account name indicated that Twitter had taken steps to ensure that th imply endorsement from Twitter, and does not mean that tweets from colloquially referred to as "blue checks" on social media and by repor
44	Twitter Zero	Twitter Zero is an initiative undertaken by social networking ser (bandwidth) charges—so-called "zero-rate"—for accessing Twitte official Twit
45	Twitterature	Twitterature (a portmanteau of Twitter and literature) is a literary use combination thereof) written by individuals or collaboratively. T
46	Unknown Number	"Unkn
47	WhyIStayed/WhyILeft	#WhyIStayed became a trending hashtag in November 2014 in defense Ravens running back, Ray Rice, punching his then-fiancee, Janay Ric hashtag was started by writer and domestic abuse survivor Beverly Go

In [67]:

```
d = []
for i in range(len(youtube_dict)):
        d.append({
                'title': list(youtube_dict.keys())[i],
                'description': list(youtube_dict.values())[i][0],
                'links': list(youtube_dict.values())[i][1],
        })
df_youtube = pd.DataFrame(d)
df_youtube['label'] = 'youtube'
df_youtube
```

Out[67]:

	title	
0	YouTube	YouTube is an American online video sharing and social media platform he Jawed Karim. It is owned by Google, and is the second most visited website, billion hours of videos each day. As of May 2019, videos were being uploa $1.65 billion. Google's ownership of YouTube expanded the site's business m$ $to offering paid content such as movies and exclusive content produced by Y$ $. YouTube also approved creators to participate in Google's AdSense progra$ 19.8 billion in 2020. In 2021, YouTube's annual advertising revenue increas apps, network television, and the ability to link with other platforms. Video ca recordings, movie trailers, teasers, live streams, vlogs, and more. M Established media corporations such as Disney, Paramount, and Warn audience.\nYouTube has had an unprecedented social impact, influencing

		YouTube has been widely criticized. Criticism of YouTube includes the
1	List of influencer boxing matches	
2	2013 YouTube Music Awards	The 2013 YouTube Music Awards, abbreviated as the YTMA, was the in streamed live from Pier 36 in New York City, with additional shows in Sec fans. The show was directed by Spike Jonze."None of us have done any
3	2015 YouTube Music Awards	The 2015 YouTube Music Awards is the second ceremony for YouT announced its intentions to reinvent the purpose of the award show. Add
4	Alternative Influence: Broadcasting the Reactionary Right on YouTube	Alternative Influence: Broadcasting the Reactionary Right on YouTube is analysis on a collection of 65
...	...	
75	YouTube Vanced	YouTube Vanced (or simply Vanced) is a discontinued, modified third-part background play, free picture-in-picture (PiP), an AMOLED black theme, sw Music version of Vanced was also developed.The name Vanced originates fro 2022, the developers of YouTube Vanced announced that the application w stop developing and distributing the app. Although the app will continue
76	Youtube-dl	youtube-dl is a free and open source download manager for video and license.As of September 2021, youtube-dl is one of the most starred proj
77	Youtubers Life	Youtubers Life is a life and business simulation video game in development
78	YouTubers vs. TikTokers	YouTubers vs. TikTokers, billed as Battle of the Platforms was a amateur bo McBroom and American TikToker Bryce Hall. The undercard included a e Rock Stadium in Miami Gardens, Florida. \nIn the main event, Austin defeated Tayler Holder in a unanimous decision. Overall, Team YouTube d two undercard matches were broadcast free of charge on social-media site
79	Template:YTMAs	Initial visibility: currently defaults to autocollapse\nTo set this template's init template collapsed, i.e., hidden apart from its title bar\n|state= {{YTMAs|state=autocollapse}}\nshows the template collapsed to the title b the template in its expanded state if there are no other collapsible items taken from the |default= parame

80 rows × 4 columns

Below are shown two ways how to access data in the dataframe. The output of both variants is identically. In SQL language the query would be performed like this:
`SELECT description FROM df_youtube WHERE title='YouTube';`

In [17]:
```
df_youtube.query("title=='YouTube'")['description']
```

Out[17]:
```
0    YouTube is an American online video sharing and social media platfo
rm headquartered in San Bruno, California. It was launched on February 1
4, 2005, by Steve Chen, Chad Hurley, and Jawed Karim. It is owned by Goo
gle, and is the second most visited website, after Google Search. YouTub
e has more than 2.5 billion monthly users who collectively watch more th
an one billion hours of videos each day. As of May 2019, videos were bei
ng uploaded at a rate of more than 500 hours of content per minute.In Oc
tober 2006, 18 months after posting its first video and 10 months after
its official launch, YouTube was bought by Google for $1.65 billion. Goo
gle's ownership of YouTube expanded the site's business model, expanding
from generating revenue from advertisements alone, to offering paid cont
ent such as movies and exclusive content produced by YouTube. It also of
fers YouTube Premium, a paid subscription option for watching content wi
thout ads. YouTube also approved creators to participate in Google's AdS
ense program, which seeks to generate more revenue for both parties. You
Tube reported revenue of $19.8 billion in 2020. In 2021, YouTube's annua
l advertising revenue increased to $28.8 billion.Since its purchase by G
oogle, YouTube has expanded beyond the core website into mobile apps, ne
twork television, and the ability to link with other platforms. Video ca
tegories on YouTube include music videos, video clips, news, short films
, feature films, documentaries, audio recordings, movie trailers, teaser
s, live streams, vlogs, and more. Most content is generated by individua
ls, including collaborations between YouTubers and corporate sponsors. E
stablished media corporations such as Disney, Paramount, and Warner Bros
. Discovery have also created and expanded their corporate YouTube chann
els to advertise to a larger audience.\nYouTube has had an unprecedented
social impact, influencing popular culture, internet trends, and creatin
g multimillionaire celebrities. Despite all its growth and success, YouT
ube has been widely criticized. Criticism of YouTube includes the websit
e being used to facilitate the spread of misinformation, copyright issue
s, routine violations of its users' privacy, enabling censorship, and en
dangering child safety and wellbeing.
Name: description, dtype: object
```

In [68]:
```
df_youtube['description'].loc[df_youtube['title']=='YouTube']
```

Out[68]:
```
0    YouTube is an American online video sharing and social media platfo
rm headquartered in San Bruno, California. It was launched on February 1
4, 2005, by Steve Chen, Chad Hurley, and Jawed Karim. It is owned by Goo
gle, and is the second most visited website, after Google Search. YouTub
e has more than 2.5 billion monthly users who collectively watch more th
an one billion hours of videos each day. As of May 2019, videos were bei
ng uploaded at a rate of more than 500 hours of content per minute.In Oc
tober 2006, 18 months after posting its first video and 10 months after
its official launch, YouTube was bought by Google for $1.65 billion. Goo
gle's ownership of YouTube expanded the site's business model, expanding
from generating revenue from advertisements alone, to offering paid cont
ent such as movies and exclusive content produced by YouTube. It also of
fers YouTube Premium, a paid subscription option for watching content wi
thout ads. YouTube also approved creators to participate in Google's AdS
ense program, which seeks to generate more revenue for both parties. You
Tube reported revenue of $19.8 billion in 2020. In 2021, YouTube's annua
l advertising revenue increased to $28.8 billion.Since its purchase by G
oogle, YouTube has expanded beyond the core website into mobile apps, ne
twork television, and the ability to link with other platforms. Video ca
tegories on YouTube include music videos, video clips, news, short films
, feature films, documentaries, audio recordings, movie trailers, teaser
s, live streams, vlogs, and more. Most content is generated by individua
ls, including collaborations between YouTubers and corporate sponsors. E
stablished media corporations such as Disney, Paramount, and Warner Bros
. Discovery have also created and expanded their corporate YouTube chann
els to advertise to a larger audience.\nYouTube has had an unprecedented
social impact, influencing popular culture, internet trends, and creatin
g multimillionaire celebrities. Despite all its growth and success, YouT
ube has been widely criticized. Criticism of YouTube includes the websit
e being used to facilitate the spread of misinformation, copyright issue
s, routine violations of its users' privacy, enabling censorship, and en
dangering child safety and wellbeing.
Name: description, dtype: object
```

Exploratory Data Analysis (EDA)

In order to gain an rough overview of the data, the data is printed and order descending where the links are the most. Justin Bieber has the most link on this article with 1132.

In [18]:
```
df_twitter.sort_values(by=['links'], ascending=False)
```

Out[18]:

	title
	Justin Drew Bieber (BEE-bər; born March 1, 1994) is a Canadian sing music. He was discovered by American record executive Scooter Braur (2009) and soon establishing himself as a teen idol.\nBieber achieved Billboard 200, making him the youngest solo male act to top the certified singles of all time in the US. His second studio album, Un experimented with dance-pop in his third studio album, Believe (2012),

4	Justin Bieber	Bieber was involved in multiple controversies and legal issues, which h won the Grammy Award for Best Dance/Electronic Recording. Th number-one singles: "Love Yourself", "Sorry", and "What Do You Mea numerous collaborations between 2016 and 2017, including "I'm the Or the first artist in history to chart new number-one songs in consecut Latin Grammy Award. In 2019, he released a country collaboration with D released his R&B-led fifth studio album, Changes, which debuted at nu US Billboard Hot 100. Bieber returned to his pop roots on his sixth st him the youngest soloist to have eight US number-one albums, a record his eighth US number-one single.Bieber is one of the best-selling m certifications from the RIAA for "Baby", "Sorry" and "Despacito". He Brit Awards, one Bambi Award, 26 Billboard Music Awards, 18 America for a male individual), and 33 Guinness World Records. Time named
33	Elon Musk	Elon Reeve Musk (EE-lon; born June 28, 1971) is a business magnate ar Inc.; founder of The Boring Company; co-founder of Neuralink and C $181 billion as of November 18, 2022, Musk is the wealthiest person in th$ $. Musk was born and grew up in Pretoria, South Africa. He attended th$ $- born mother. Two years later, he matriculated at Queen's Universit$ $where he received bachelor's degrees in economics and physics. He mov$ $but dropped out after two days to instead pursue a business career, co -$ 307 million in 1999. The same year, Musk co-f $1.5 billion. \nIn 2002, Musk founded SpaceX, an aerospace manu fact$ $he was an early investor in the electric vehicle manu facturer Tesla Mo$ $eventually assuming the position of CEO in 2008. In 2006, he helped cre$ $- founded OpenAI, an non profit artificial intelligence research comp$ $- computer interfaces — and The Boring Company, a tunnel construc$ 44 billion. He has proposed a hyperloop high-speed vactrain transpor regarding politics and technology, particularly on Twitter, and is a highl COVID-19 misinformation and promoting a conspiracy theory concer for falsely tweeting that he had secured funding for a private takeover the SEC. Following his failed submarine proposal during the Tha
0	Twitter	Twitter is a microblogging, social networking service owned by Ameri post, like, and retweet tweets, while unregistered users or programmatically via its APIs. Before April 2020, services were a languages in November 2017. Audio and video tweets remain limited t March 2006 and launched in July of that year. Twitter, Inc. is based in S 340 million tweets a day, and the service handled an average of 1.6 billi of the Internet". By the start of 2019, Twitter had more than 330 mi accounts were estimated to make up 15% of Twitter accounts, appro Musk, making it one of the biggest deals to take a company private. Twitter has been cr
17	2009 Iranian presidential election protests	After incumbent president Mahmoud Ahmadinejad declared v candidates Mir-Hossein Mousavi and Mehdi Karroubi. The pro proponents, reflecting Mousavi's campaign theme, and Persian Awak that incumbent President Mahmoud Ahmadinejad won nearly 63 percent manipulated and the election was rigged, with Rezaee and Mousavi appeal against the result to the Guardian Council on 14 June. Ayato Movement leaders. Ahmadinejad called the election "completely free" a match".Due to the sporadic cases of violence present at the pro sprayed, arrested and tortured, and even shot in some cases. The mos around the world. Opposition groups also reported thousands more w children by the Islamic Revolutionary Guards in prisons such as K reports by Mousavi supporters allege at least 72 deaths (over t documents citing death by heart attack or meningitis. Iranian authorities

		Green Movement developed during these protests. It was
14	Hashtag	A hashtag is a metadata tag that is prefaced by the hash (also know such as Twitter or Instagram as a form of user-generated tagging #bluesky returns all posts that have been tagged with that term proposed by American blogger and product consultant Chris Messina i owned by no one". Hashtags became entrenched in the culture of T English Dictionary as "a word or phrase with the symbol # i
18	ISIS-chan	ISIS-chan (ISIS（アイシス）ちゃん, aishisuchan) is a fictional man incident
37	Twitter trends	On Twitter, a word, phrase, or topic that is mentioned at a greater concerted effort by users or because of an event that prompts people and what people's opinions are about it.Trending topics are sometim
42	Twitter usage	Since the
34	Stan Twitter	Stan Twitter is a community of Twitter users that post opinions relate terminology but also for incidents of harassm
10	Chaos with Ed Miliband	"Chaos with Ed Miliband" is a phrase from a 2015 tweet by David Cam before the 2015 United Kingdom general election – Cameron, the l Government" with him, or "chaos with Ed Miliband", who was leadin
6	Black Twitter	Black Twitter is an internet community largely consisting of African-Am Feminista Jones described it in Salon as "a collective of active, primari
15	History of Twitter	
20	List of Twitter features	Twitter is an American microblogging and social networking service on v
5	Billboard Twitter Real-Time	The Billboard Twitter Real-Time charts were four interactive Billboard mu based on how often they are mentioned in messages called "tweets" se Tracks" and "Weekly Emerging Artists" were reported to define how far U.S. The first two charts updated on a "real-time" minute-by-minu "Trending 140" and real-time charts were discontinued and the Emerg rising acts of each week.In October 2021, Billboard and Twitter launch first ch
30	Shorty Awards	The Shorty Awards, also known as the "Shortys", is an annual awards YouTube, Instagram, TikTok, Twitch and the rest of the social web.\nThe recognized content creation on other social networking sites, including live. Each award recognizes the entire body of work of the content cr commented: "The Shortys recognize that social media is about more tha
45	Twitterature	Twitterature (a portmanteau of Twitter and literature) is a literary use combination thereof) written by individuals or collaboratively. T
39	Twitter diplomacy	Twitter diplomacy, also called "Twiplomacy" or "hashtag diplomacy," is t their diplomats to conduct diplomatic outreach and public dip
13	Frogtwitter	#Frogtwitter is a network of loosely-connected pseudonymous Twi could also deri
36	Tawawa on Monday	Tawawa on Monday (Japanese: 月曜日のたわわ, Hepburn: Getsuyōb Himura has posted an illustration on his Twitter account every

		December 2016. A second season by Yokohama Animation Labora
12	FlockDB	FlockDB is an open-source distributed, fault-tolerant graph database e.g. followings and favorites. FlockDB differs from other graph database primary use-case for Redis sets. Since it is still in the process of bein FlockDB was posted on GitHub shortly after Twitter released its Gizza
43	Twitter verification	Twitter verification is a system intended to communicate the auth distinguish genuine notable account holders, such as celebrities and org account name indicated that Twitter had taken steps to ensure that th imply endorsement from Twitter, and does not mean that tweets from colloquially referred to as "blue checks" on social media and by repor
38	Twitter bomb	The term twitter bomb or tweet bomb (also spelled as one word) refers @messages, from multiple accounts, with the goal of advertising a certa
25	Republik Twitter	Republik Twitter (stylised as #republiktwitter; meaning The Twitter Rep Aryasatya, it follows a university student in his quest to be with a beaut script. Based on one of his earlier blog posts, pre-production for the
7	Bluesky (protocol)	Bluesky is an initiative to develop a decentralized social network prot Bluesky
22	Mention (blogging)	A mention (also known as @replies or tagging) is a means by which a attention to) another user of a social networking or blogging serv
19	KHive	#KHive is the hashtag used by an informal online community supporting and occasionally transcribed as K-Hive, and refers to the wider online 2020 presidential campaign as an effort to defend Harris from racist
8	Bootstrap (front-end framework)	Bootstrap is a free and open-source CSS framework directed at re templates for typography, forms, buttons, navigation, and other inter
28	Safe Space (South Park)	"Safe Space" is the fifth episode of the nineteenth season and the 262n The episode premiered on Comedy Central on October
31	Skank Hunt	"Skank Hunt" is the second episode in the twentieth season of the Ame
47	WhyIStayed/WhyILeft	#WhyIStayed became a trending hashtag in November 2014 in defense Ravens running back, Ray Rice, punching his then-fiancee, Janay Ric hashtag was started by writer and domestic abuse survivor Beverly Go
16	Internet activism during the 2009 Iranian election protests	Internet activism and, specifically, social networking has been instrume video, and Twitter, Facebook, and blogs have been places for prot organi
41	Twitter Revolution	Twitter Revolution is a term used to refer to different revolutions and pro
1	Aufschrei	#Aufschrei ("outcry") is a German hashtag which went viral on the s Tweets began to appear in response to the publication of an article in w Democratic Party (FDP); within
		Twitter Zero is an initiative undertaken by social networking serv

44	Twitter Zero	(bandwidth) charges—so-called "zero-rate"—for accessing Twitte official Twit
24	Museum Week	Museum Week is a global event that takes place during one week each collaboration of twelve French museums wishing to expand their audie
40	Twitter Power	Twitter Power: How to Dominate Your Market One Tweet at a Time Twitter account, market their brands, and gain Twitter followers.\nPu giving r
21	The Magical Moment	The Magical Moment is a TweetBook by Brazilian author Paulo Coelho.
9	Breaking Tweets	Breaking Tweets was an international journalism news Web site that tra "hyperlocal gone global," personalizing major news events throug Breaking Tweets Sports, Breaking Tweets Entertainment, and Br
11	Dogs at polling stations	Dogs at polling stations or #dogsatpollingstations is a popular hashtag dogs are photographed waiting for their owners outside the polling s general elections of 2015 and 2017.The 2019 United Kingdom gener wearing a Santa hat. Politicians who participated included Boris Johns Carrot, as they do not have a dog. Other animals, such as horses, also
2	100 Nichi Go ni Shinu Wani	100 Nichi Go ni Shinu Wani (100日後に死ぬワニ, "A Crocodile Who Will a crocodile's last 100 days before his death, was self-published on Twi 8, 2020. An anime film adaptation produced by TIA, titled 100 Nichik
29	Takahiro Shiraishi	Takahiro Shiraishi (白石隆浩, Shiraishi Takahiro, born 9 October 1990) reports at the time of his sentencing. In Zama, Japan,
35	Such Tweet Sorrow	Such Tweet Sorrow was a modern-day adaptation of Shakespear performed the play on Twitter and other web devices. The actors imp comments via T
3	Artificial Intelligence for Digital Response	Artificial Intelligence for Digital Response (AIDR) is a free and open sou It has been developed by the Qatar Computing Research Institute and his team "have developed novel computational techniques and technolog - according to him the system "combines human intelligence
32	Snowflake ID	Snowflake IDs, or snowflakes, are a form of unique identifier used a
23	Mikeyy	Mikeyy is the name of a computer worm that spread approximately discrete attacks "between 2 AM Saturday April 11, 2009 Pacific tim called StalkDaily.The worm was written by 17-year-old Michael Moon
46	Unknown Number	"Unkn
27	Rotation Curation	Rotation Curation, also #RotationCuration, is the concept of rotating the a group,
26	Roger Cosmonkey	Roger Cosmonkey is a series of webcomics created by filmmaker Cory E Cosmonkey 2: Mega-Pigeon.\nBoth series were posted on Edwards' included a link to a single comic strip in the series.\nIt has been describ feature film of the television series Fraggle Rock expired. He fou

In [19]:
```
df_youtube.sort_values(by=['links'], ascending=False)
```

Out[19]:

	title	
0	YouTube	YouTube is an American online video sharing and social media platform headq Jawed Karim. It is owned by Google, and is the second most visited website, aft billion hours of videos each day. As of May 2019, videos were being uploadec $1.65 billion. Google's ownership of YouTube expanded the site's business mode to offering paid content such as movies and exclusive content produced by YouT . YouTube also approved creators to participate in Google's AdSense program, 19.8 billion in 2020. In 2021, YouTube's annual advertising revenue increased t apps, network television, and the ability to link with other platforms. Video categ recordings, movie trailers, teasers, live streams, vlogs, and more. Most Established media corporations such as Disney, Paramount, and Warner E audience.\nYouTube has had an unprecedented social impact, influencing pc YouTube has been widely criticized. Criticism of YouTube includes the web
48	Vevo	Vevo (Vee-vo, an abbreviation for "Video Evolution", stylized as VEVO until 20 The service is also available as an app on selected smart TVs, digital vide mobile and tablet app, however this was shut down in May 2018 to allow the se joint venture among three major record companies: Universal Music Group (UN Group (WMG), the world's third-largest record company, agreed to license p Group and Sony Music Entertainment, syndicated on YouTube and its app, and be considering hosting its content on the service after it launched, but forme renewed interest in licensing music from WMG and a deal with WMG was comple
73	YouTube Theater	YouTube Theater is a 6,000 seat music and theater venue in Inglewoo entertainment com
51	Susan Wojcicki	Susan Diane Wojcicki (wuu-CHITS-kee; born July 5, 1968) is a Polis years.Wojcicki was involved in the founding of Google, and became Google' charge of Google's original video service. After observing the success of YouT
69	YouTube Shorts	YouTube Shorts is a short-form video-sharing platform offered by YouTube
...	...	
19	InfoChammel	InfoChammel is a Youtube channel and webseries that satirizes traditional televi teletext services and infomercials. Since beginning as an intermittent web
25	Keeping It Real Art Critics	Keeping It Real Art Critics (also known as KIRAC) is a web series on YouTube a KIRAC episodes occupy the middle ground between art and art criticism and fo the platform has gained both praise and notoriety as a result of its confrontati Ruitenbeek's personal sphere. Contributors to this dialogue, whether friends or archaeologist and philosopher Kate Sinha
55	YouTomb	YouTomb was a website built to track videos removed by popular video-sl tracked not only DMCA takedowns but also terms of use violations and user database was generated by software that repeatedly scanned YouTube for unav is licensed under the GNU Affero General Public License. Although it only tr
44	The Slot (TV series)	The Slot is an Australian sketch comedy series produced for The Come

38	Petit Tube	Petit Tube is a French website that searches through an algorithm for obscure to view videos that would have otherwise been seen by hardly any people.\

80 rows × 4 columns

Drop missing values

The YouTube-dataframe contains two missing descriptions. The code below drops the rows with missing description. The code 'searches' all description where the input is empty. Then it gets the index, and finally drops them.

In [20]:
```
df_youtube.drop(df_youtube.loc[df_youtube['description']==''].index, inpl
```

Adding features to the data frame

For every title and description the length is counted in characters, these values are inserted into the data frame. Additionally, are the description token counted. The same procedure is performed for Twitter and YouTube as before.

In [70]:
```
df_twitter['title length'] = df_twitter['title'].astype(str).apply(len)
df_twitter['description length'] = df_twitter['description'].astype(str).
df_twitter['description token'] = df_twitter['description'].apply(lambda
df_twitter
```

Out[70]:

	title	
0	Twitter	Twitter is a microblogging, social networking service owned by Americar "tweets". Registered users can post, like, and retweet tweets, while Twitter through browser or mobile frontend software, or programmatic originally restricted to 140 characters, but the limit was doubled to 28(to 140 seconds for most accounts.\nTwitter was created by Jack Dorse that year. Twitter, Inc. is based in San Francisco, California and has m 340 million tweets a day, and the service handled an average of 1.6 billic has been described as "the SMS of the Internet". By the start of majority of tweets are written by a minority of users. As of 2020, fak million accounts.On April 25, 2022, the Twitter board of directors agreec company private. After a protracted period of controversy and legal b criticized for increasingly promoting
1	Aufschrei	#Aufschrei ("outcry") is a German hashtag which went viral on experiences of sexism in Germany. The Tweets began to appea describes an invasive encounter with politician Rainer Brüderle c tr
2	100 Nichi Go ni Shinu Wani	100 Nichi Go ni Shinu Wani (100日後に死ぬワニ, "A Crocodile Who Will Kikuchi. The series, which depicts a crocodile's last 100 days before his 2020. A manga compilation was published by Shougakukan on April 8, 2(日間生きたワニ, "A Crocodile Who Lived For 100 Days"), was sche

3	Artificial Intelligence for Digital Response	Artificial Intelligence for Digital Response (AIDR) is a free and open so disasters, and humanitarian crises. It has been developed by the Qa Source Software World Challenge.Muhammad Imran stated that he which can help gain insightful and actionable information from online so human intelligence with machine learning techniques,
4	Justin Bieber	Justin Drew Bieber (BEE-bər; born March 1, 1994) is a Canadian s influential role in modern-day popular music. He was discovered by A gaining recognition with the release of his debut seven-track E commercial success with his teen pop-driven debut studio albu youngest solo male act to top the chart in 47 years. The album spaw certified singles of all time in the US. His second studio album, Under t number one in the US. Bieber experimented with dance-pop in his third five US number-one albums at age 18. Between 2013 and 2014, Bieber w on his public image.In 2015, Bieber explored EDM with the releas Recording. The song's musical direction played an instrumental role i "Love Yourself", "Sorry", and "What Do You Mean?". He became the diversified on numerous collaborations between 2016 and 2017, includi US Billboard Hot 100 a week apart, making him the first artist in histo the greatest Latin song of all time by Billboard and earned Bieber his Dan + Shay, "10,000 Hours", which won the Grammy Award for Best album, Changes, which debuted at number one in both the UK and US, a US Billboard Hot 100. Bieber returned to his pop roots on his sixth s atop the US Billboard 200, making him the youngest soloist to have e The same year, Bieber released "Stay" with the Kid Laroi, which became all time, with estimated sales of over 150 million records worldwide. He i "Despacito". He has received numerous accolades, including two Gr Bambi Award, 26 Billboard Music Awards, 18 American Music Awards, 22 (the most wins for a male individual), and 33 Guinness World Records. T he was includ
5	Billboard Twitter Real-Time	The Billboard Twitter Real-Time charts were four interactive Billboard m from popular and emerging artists based on how often they are m United States. The charts "Trending 140", "Emerging Artists", "We interacted with, and influenced, popular content by ranking the most p "real-time" minute-by-minute basis, with the last two charts providing w 140" and real-time charts were discontinued and the Emergin highlighting the top-performing rising acts of each week.In Octobe Time" - first 24-hours started at noon on October 22, and "Weekly" -
6	Black Twitter	Black Twitter is an internet community largely consisting of African black community in the United States. Feminista Jones described it in S created a virtual community ... [and are] proving adept at bringing abou
7	Bluesky (protocol)	Bluesky is an initiative to develop a decentralized social network protoc in a research phase as of 2022. Bluesky is owned by the team itse
8	Bootstrap (front-end framework)	Bootstrap is a free and open-source CSS framework directed a (optionally) JavaScript-based design templates for typography, forms,
9	Breaking Tweets	Breaking Tweets was an international journalism news Web site th Internet. It considered its content to be "hyperlocal gone global," per launching on January 31, 2009, the site launched three spin-off sit Chicago. The main site, the world version, received visit
10	Chaos with Ed Miliband	"Chaos with Ed Miliband" is a phrase from a 2015 tweet by David Ca Opposition. On 4 May – a few days before the 2015 United Kingdom that the British public faced a choice between "stability and strong Gov

		into the election. The political turmoil in the United Kingdom in the wake
11	Dogs at polling stations	Dogs at polling stations or #dogsatpollingstations is a popular countries such as Australia. Typically, the dogs are photographed v services such as Instagram or Twitter.The hashtag became popul election was held in December and thus many of the photographs ha participated included Boris Johnson, with his dog Dilyn, and Sadiq K Carrot, as they do not have a dog. Other animals, such as horses, also
12	FlockDB	FlockDB is an open-source distributed, fault-tolerant graph databas store relationships between users, e.g. followings and favorites. FlockDB hop graph traversal but rather for rapid set operations, not unlike the pr outside of Twitter use, the code is still very rough and hence there is r released its Gizzard framework, which it uses to query the FlockDB dis
13	Frogtwitter	#Frogtwitter is a network of loosely-connected pseudonymous Twitte Josh Vandiver speculates that it could also derive from Aristophanes
14	Hashtag	A hashtag is a metadata tag that is prefaced by the hash (a microblogging and photo-sharing services such as Twitter or Instagram topic or theme. For example, a search within Instagram for the hashta hash symbol, a hashtag may include letters, numerals, or unde consultant Chris Messina in a 2007 tweet. Messina made no attempt to no one". Hashtags became entrenched in the culture of Twitter and soc added to the Oxford English Dictionary as "a word or phrase with t
15	History of Twitter	
16	Internet activism during the 2009 Iranian election protests	Internet activism and, specifically, social networking has been instru been uploading amateur pictures and video, and Twitter, Faceboo Although some scholars in the West stress that Twitter has been used t
17	2009 Iranian presidential election protests	After incumbent president Mahmoud Ahmadinejad declared victory in th in support of opposition candidates Mir-Hossein Mousavi and M Movement (Persian: جنبش سبز Jonbesh-e Sabz) by their proponents Green Revolution.Protests began on the night of 12 June 2009, followin 63 percent of the vote, despite several reported irregularities. Hov election was rigged, with Rezaee and Mousavi lodging official complai an official appeal against the result to the Guardian Council on 14 June and irregularities, as requested by Green Movement leaders. Ahmadineja dismissing the protests as akin to little more than "passions after a government had the police and paramilitary Basij violently suppress the in some cases. The most widely known firearm victim was Neda Agha- world. Opposition groups also reported thousands more were arrested a of men, women, and children by the Islamic Revolutionary Guards in pr 36 people during the protests, while unconfirmed reports by Mou following the election. They claimed relatives of the deceased were force closed universities in Tehran, blocked websites, cut off mobile signals these protests. It was also termed the "Twitter Revoluti
18	ISIS-chan	ISIS-chan (ISIS（アイシス）ちゃん, aishisuchan) is a fictional manga created in 2015 after a series of incidents in which Japanese nationals
		#KHive is the hashtag used by an informal online community supporting term which is always pronounced and occasionally transcribed as K-Hi

19	KHive	campaign or office. The community formed prior to and during her 2020 and misinformation. Th
20	List of Twitter features	Twitter is an American microblogging and social networking service on can post, like, and retw
21	The Magical Moment	The Magical Moment is a TweetBook by Brazilian author Paulo Coelh those years in Twitter. Illustrated by Joong Hwan Hwang, it was origina
22	Mention (blogging)	A mention (also known as @replies or tagging) is a means by which getting the attention of (or drawing attention to) another user of a soci or as
23	Mikeyy	Mikeyy is the name of a computer worm that spread appro microblogging website Twitter.com in four discrete attacks "bet morning" before it was "identified and deleted". The tweets promoted who operates a website to point
24	Museum Week	Museum Week is a global event that takes place during institutions.Museum Week started in 2014 with a collaboration of twelve a different international cause.In 2015, tl
25	Republik Twitter	Republik Twitter (stylised as #republiktwitter; meaning The Twitter Re Starring Laura Basuki and Abimana Aryasatya, it follows a university stu in June 2011, when screenwriter Eddri Sumitra Ito began preparing a sc months, with another fourteen days shooting and four months post-pro
26	Roger Cosmonkey	Roger Cosmonkey is a series of webcomics created by filmmaker Cory series; Roger Cosmonkey and Roger Cosmonkey 2: Mega-Pigeon.\nBo day, over the course of ten days. Each tweet, called a "Twittersode" world's first episodic Twitter series.Edwards developed the idea sh Rock expired. He found the process of creating the series for Twitter t
27	Rotation Curation	Rotation Curation, also #RotationCuration, is the concept of rotating t location, a country, an organization, a group, and so on. The concept is
28	Safe Space (South Park)	"Safe Space" is the fifth episode of the nineteenth season and tl directed by series co-creator Trey Parker. The episode premiered on C
29	Takahiro Shiraishi	Takahiro Shiraishi (白石隆浩, Shiraishi Takahiro, born 9 October 1990) i he was labeled as in most media reports at the time of his sentenc
30	Shorty Awards	The Shorty Awards, also known as the "Shortys", is an annual awards content across Twitter, Facebook, YouTube, Instagram, TikTok, Twitch for achievements on the Twitter platform. Since then, the awards ha Instagram, Vine, Snapchat, YouNow, and Periscope.The awards ceremor work of the content creator for the calendar year, rather than an indivi "The Shortys recognize that social media is about more than who can
31	Skank Hunt	"Skank Hunt" is the second episode in the twentieth season of the overall, it first aired on Comedy Central in the United States on Septer
32	Snowflake ID	Snowflake IDs, or snowflakes, are a form of unique identifier used in d tweets. The format has been adopted by other companies, includ
		Elon Reeve Musk (EE-lon; born June 28, 1971) is a business magnate CEO and product architect of Tesla, Inc.; founder of The Boring Com

33	Elon Musk	$181 billion as of November 18, 2022, Musk is the wealthiest person in the$ $- time billionaires list. Musk was born and grew up in Pretoria, South$ $. He attended the University of Pretoria before moving to Canada at age$ $. Two years later, he matriculated at Queen's University and transferr$ $where he received bachelor's degrees in economics and physics. He mov$ $but dropped out after two days to instead pursue a business career, co-$ $the startup was acquired by Compaq for$ 307 million in 1999. The same year, Musk co-founded the online bank X $1.5 billion. \nIn 2002, Musk founded SpaceX, an aerospace manufact$ $. In 2004, he was an early investor in the electric vehicle manufacturer T$ $. He became its chairman and product architect, eventually assuming t$ $a solar energy company that was later acquired by Tesla and became Tes$ $a nonprofit artificial intelligence research company. The following y$ $— a neurotechnology company developing brain-computer interfaces$ $Musk purchased the social media platform Twitter for$ 44 billion. He has proposed a hyperloop high-speed vactrain transportat made controversial statements regarding politics and technology, part making unscientific and misleading statements, such as spreading COV a US politician's husband. In 2018, the U.S. Securities and Exchange C private takeover of Tesla. Musk stepped down as chairman of Tesla and his failed submarine proposal during the Tham Luang cave rescue,
34	Stan Twitter	Stan Twitter is a community of Twitter users that post opinions rela been noted for its particular shared terminology but also for incidents of
35	Such Tweet Sorrow	Such Tweet Sorrow was a modern-day adaptation of Shakespeare's R 2010) six professional actors performed the play on Twitter and other w with each other and react to followers, fans, real events and comments
36	Tawawa on Monday	Tawawa on Monday (Japanese: 月曜日のたわわ, Hepburn: Getsuyōb Himura (比村奇石, Himura Kiseki). Himura has posted an illustratior animation (ONA) adaptation by Pine Jam aired from October to September to December 2021. A m
37	Twitter trends	On Twitter, a word, phrase, or topic that is mentioned at a greater r become popular either through a concerted effort by users or becaus Twitter and their users to understand what is happening in the world a concerted efforts and manipulations by fans of certain celebrities or
38	Twitter bomb	The term twitter bomb or tweet bomb (also spelled as one word) refe and other similar content, including @messages, from multiple acco feeds with the same message, and making it a "trendi
39	Twitter diplomacy	Twitter diplomacy, also called "Twiplomacy" or "hashtag diplo intergovernmental organizations (IGOs), and their diplomats to diplomatic communications, ranging from cordial announcements of b
40	Twitter Power	Twitter Power: How to Dominate Your Market One Tweet at a Time Twitter, explaining how to create a Twitter account, market their brand received positive reviews. It was criticized by a CNET reviewer for gi

41	Twitter Revolution	Twitter Revolution is a term used to refer to different revolutions an
42	Twitter usage	Since the launch of Twitter on July 15, 2006
43	Twitter verification	Twitter verification is a system intended to communicate the authenti site's readers with a means to distinguish genuine notable accour November 2022 a blue checkmark (UK: blue tick) displayed against a was actually owned by the person or organization whom it claimed to r mean that tweets from a verified account are necessarily accurate or referred to as "blue checks" on social media and by reporters. Since N
44	Twitter Zero	Twitter Zero is an initiative undertaken by social networking serv whereby the providers waive data (bandwidth) charges—so-called "z version of the website. The images could be loaded by using the offic
45	Twitterature	Twitterature (a portmanteau of Twitter and literature) is a literar aphorisms, poetry, and fiction (or some combination thereof) writ r
46	Unknown Number	"Unknown Number" is a science fiction s
47	WhyIStayed/WhyILeft	#WhyIStayed became a trending hashtag in November 2014 in defens appeared to show former Baltimore Ravens running back, Ray Rice, pun other victims of abuse choose to stay in abusive relationships. The hash in an effort to "change the tone of the conversation." It began to tren

In [71]:
```
df_youtube['title length'] = df_youtube['title'].astype(str).apply(len)
df_youtube['description length'] = df_youtube['description'].astype(str).
df_youtube['description token'] = df_youtube['description'].apply(lambda
df_youtube
```

Out[71]:

	title	
0	YouTube	YouTube is an American online video sharing and social media platform headc Steve Chen, Chad Hurley, and Jawed Karim. It is owned by Google, and is the 2.5 billion monthly users who collectively watch more than one billion hours c more than 500 hours of content per minute.In October 2006, 18 months afte $1.65 billion. Google's ownership of YouTube expanded the site's business m too offering paid content such as movies and exclusive content produced by Y a paid subscription option for watching content without ads. YouTube also c which seeks to generate more revenue for both parties. YouTube reported re 19.8 billion in 2020. In 2021, YouTube's annual advertising revenue incr beyond the core website into mobile apps, network television, and the abi videos, video clips, news, short films, feature films, documentaries, a content is generated by individuals, including collaborations between Disney, Paramount, and Warner Bros. Discovery have also crea audience.\nYouTube has had an unprecedented social impact, influenci Despite all its growth and success, YouTube has been widely criticized. Crit misinformation, copyright issues, routine violations of its users
	List of influencer	

1	boxing matches	These bout
2	2013 YouTube Music Awards	The 2013 YouTube Music Awards, abbreviated as the YTMA, was the inaugur held on November 3, 2013, streamed live from Pier 36 in New York City, wit other award shows, the winners were entirely voted on by fans. The show wa awards show – in a way we're all like amateurs on YouTube ourselves,
3	2015 YouTube Music Awards	The 2015 YouTube Music Awards is the second ceremony for YouTube Mu November 2014, YouTube announced its intentions to reinvent the purp cerem
4	Alternative Influence: Broadcasting the Reactionary Right on YouTube	Alternative Influence: Broadcasting the Reactionary Right on YouTube is a 2 Society that performs network analysis on a collection of 65 political influ
...	...	
75	YouTube Vanced	YouTube Vanced (or simply Vanced) is a discontinued, modified third-party the app include SponsorBlock, background play, free picture-in-picture (P the ability to restore dislike counts on YouTube videos. A YouTube Music v word advanced but with ad removed, in reference to the software's a announced that the application would be shut down after they received developing and distributing the app. Although the app will continue to functi
76	Youtube-dl	youtube-dl is a free and open source download manager for video and auc under the Unlicense software license.As of September 2021, youtub According to libraries.io, 308 other packages and 1.43k repositories dep
77	Youtubers Life	Youtubers Life is a life and business simulation video game in develo
78	YouTubers vs. TikTokers	YouTubers vs. TikTokers, billed as Battle of the Platforms was a amateur between American YouTuber Austin McBroom and American TikToker Bryce H and TikToker Tayler Holder. It was held on June 12, 2021 at Hard Rock Stadiu Bryce Hall in three rounds with a technical knockout. In the penultimate event YouTube defeated Team TikTok 6-1. \nThe event was promoted as the Bat broadcast free of charge on social-media sites, and the pay-per-view was b
79	Template:YTMAs	Initial visibility: currently defaults to autocollapse\nTo set this templat {{YTMAs|state=collapsed}} to show the template collapsed, i.e., hidden apart template expanded, i.e., fully visible\n|state=autocollapse: {{YTMAs|stat {{navbox}}, a {{sidebar}}, or some other table on the page with the collapsi collapsible items on the page\nIf the |state= parameter in the template on t parameter in the Collapsible option templ

80 rows × 7 columns

When applying `.describe()` to a data frame a statistical overview of the data is shown. As we can see below does it contain the count, mean value, standart deviation, minimum and maximum and the percentiles. Of course only numerical values can be displayed in this way, therefore the title and description are not listet below. The YouTube dataframe contains 80 observations and the Twitter dataframe only 48. The title and description length and the description token are similar for both data sets. However, the YouTube dataset contains in average more links than the Twitter dataset.

In [74]:
```
df_twitter.describe()
```

Out[74]:

	links	title length	description length	description token
count	48.000000	48.000000	48.000000	48.000000
mean	186.437500	17.229167	688.145833	108.833333
std	266.868595	10.139956	686.712396	109.726417
min	8.000000	5.000000	77.000000	14.000000
25%	31.000000	12.000000	341.750000	51.750000
50%	83.500000	15.000000	490.500000	77.500000
75%	216.000000	18.250000	680.000000	108.500000
max	1132.000000	59.000000	3520.000000	584.000000

In [75]:
```
df_youtube.describe()
```

Out[75]:

	links	title length	description length	description token
count	80.000000	80.000000	80.000000	80.000000
mean	203.675000	18.825000	657.712500	102.150000
std	314.559961	10.962277	607.269656	91.043612
min	12.000000	4.000000	0.000000	0.000000
25%	31.750000	11.750000	310.500000	48.500000
50%	119.000000	16.500000	486.000000	75.500000
75%	168.750000	23.250000	800.500000	126.500000
max	1745.000000	68.000000	3376.000000	513.000000

Word Cloud

In this step are all descriptions for Twitter and YouTube aggregated to a single string for each topic. The strings are saved into the variables `twitter_desc` and `youtube_desc`

In [76]:
```
twitter_desc = df_twitter['description'].str.cat()
youtube_desc = df_youtube['description'].str.cat()
```

A wordcloud is made for each topic. The top tags for Twitter are Twitter, tweet, Bieber, hashtag, Musk. All this keyword make absolutely sense. For YouTube the top tags are YouTube, Video, content, google, channel, music also in this case it makes sense.

In [78]:
```
rows=1
cols=2

row=0
col=0
titles = ['Twitter', 'YouTube']
w = [twitter_desc, youtube_desc]
fig, ax = plt.subplots(1, 2, figsize=(12.5,6.5))

masks = [np.array(Image.open('twitter-mask.jpg')), # if masks file not av
                 np.array(Image.open('youtube-mask.jpg'))] # if masks fil

for i in range(2):
  ax[col].imshow(WordCloud(width = 500, height = 500,
                 background_color ='white',
                                 mask=masks[i], # if masks file not availa
                 min_font_size = 10).generate(w[i]))
  ax[col].axis("off")
  ax[col].set_title(titles[i], fontdict={'fontsize': 12, 'fontweight' : '
  row=row+1
  if row==rows:
     row=0
     col=col+1

plt.subplots_adjust(left=0.07, right=0.93, wspace=0.1, hspace=0.0,top=0.9
plt.show()
```

Twitter

Twitter
tweet
Musk
website
serviceseries
Award
day
first
use
became
one
year
company
Bieber
world
post
user
social
media
account
protests
hashtag
community
chart
May
around
two
used
site
Iranian
Internet
born
artist
event
CEO
trending
web
election
president
October
posted
public
blue
University
real time
episode
Tesla top
written
published
character
based
message
via
UK
Instagram
November
Inc
single
billion
million
online
new
including
network
time
form
organization
record
American
popular
April
June
released
filled
week
Billboard
many
people
music
received
dog
aikoblogging
Spanish
Japanese
created
following
social networking
weekly
known
making
three
number one
history
Grammy
Oscars
Emmy
Grammys
Boston
Twitter's
Followers
Blue
release
Twitter's
Tweet
Twitter's
Tweets
Twitter's
Twitter's
Twitter's

Out[84]:

	links	title length	description length	description token	polarity
count	48.000000	48.000000	48.000000	48.000000	48.000000
mean	186.437500	17.229167	688.145833	108.833333	0.091598
std	266.868595	10.139956	686.712396	109.726417	0.118560
min	8.000000	5.000000	77.000000	14.000000	-0.088889
25%	31.000000	12.000000	341.750000	51.750000	0.011806
50%	83.500000	15.000000	490.500000	77.500000	0.072917
75%	216.000000	18.250000	680.000000	108.500000	0.149980
max	1132.000000	59.000000	3520.000000	584.000000	0.500000

In [85]:

```
df_youtube.describe()
```

Out[85]:

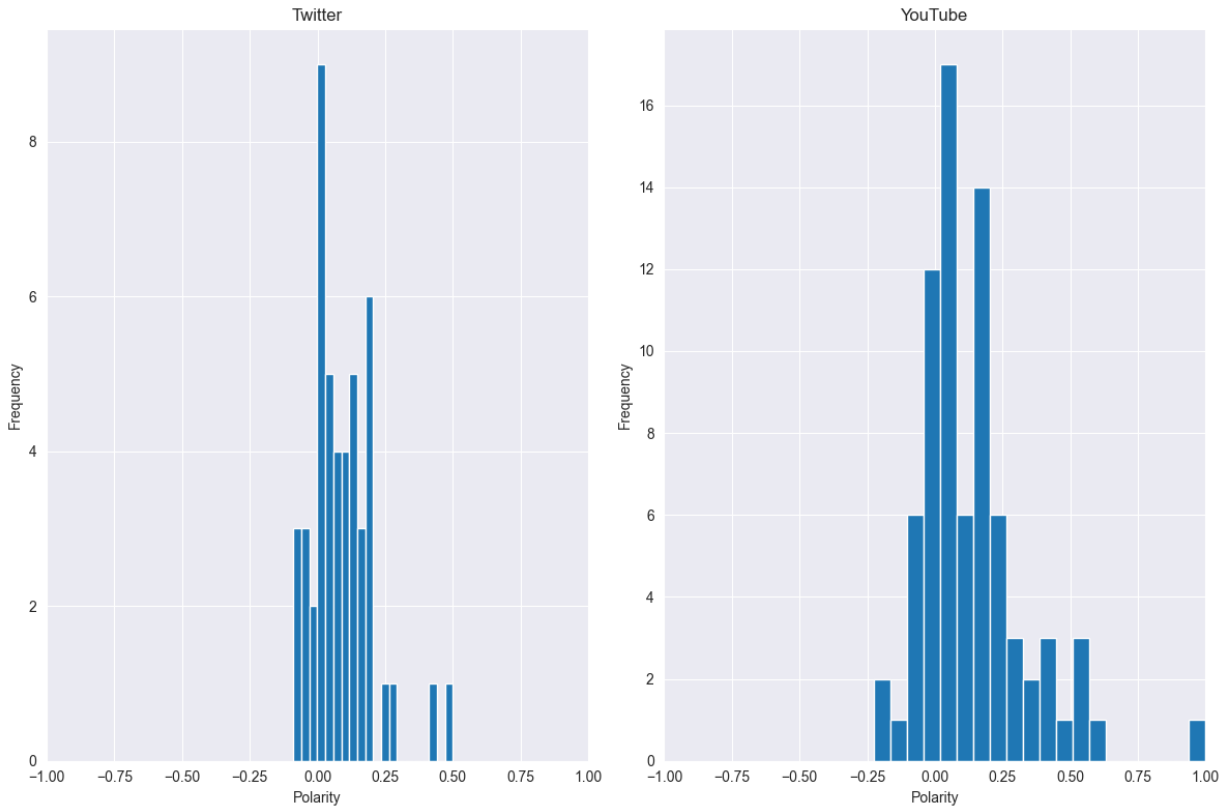
	links	title length	description length	description token	polarity
count	80.000000	80.000000	80.000000	80.000000	80.000000
mean	203.675000	18.825000	657.712500	102.150000	0.134878
std	314.559961	10.962277	607.269656	91.043612	0.194517
min	12.000000	4.000000	0.000000	0.000000	-0.225379
25%	31.750000	11.750000	310.500000	48.500000	0.000000
50%	119.000000	16.500000	486.000000	75.500000	0.085795
75%	168.750000	23.250000	800.500000	126.500000	0.203598
max	1745.000000	68.000000	3376.000000	513.000000	1.000000

Graphical analysis

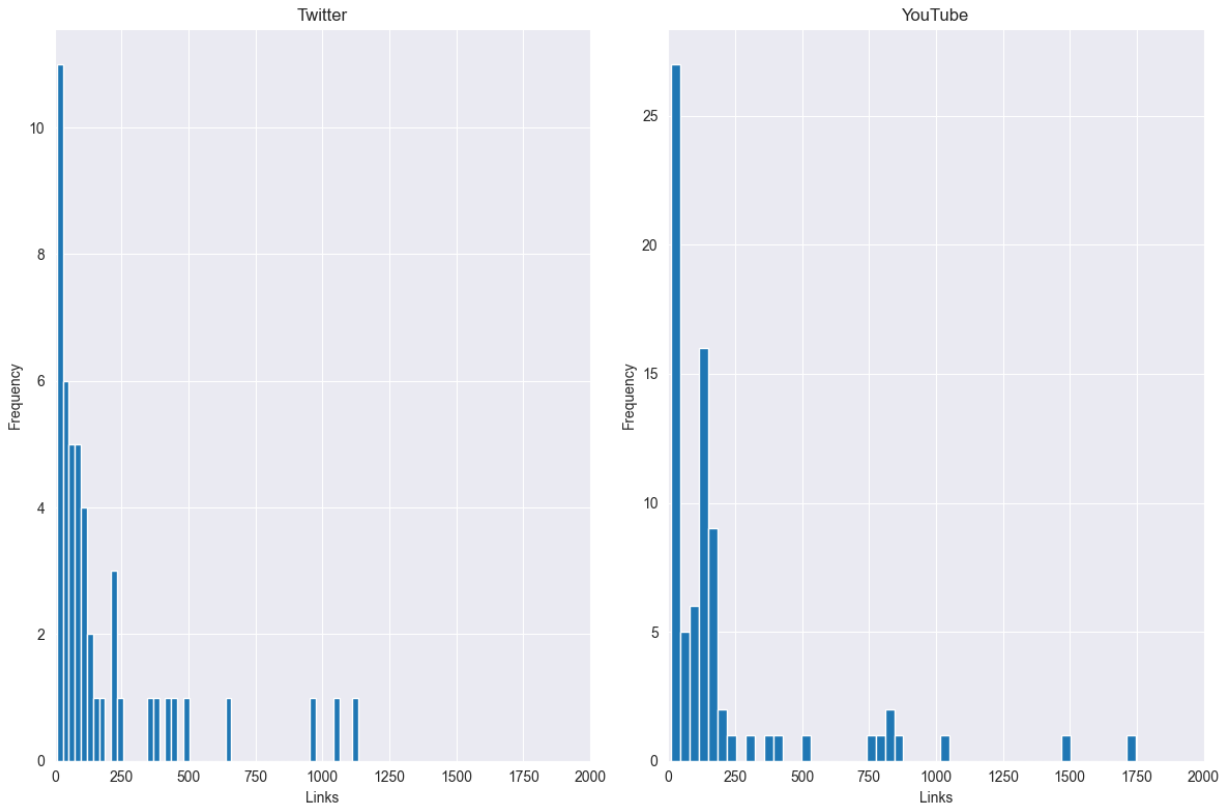
The following section shows the graphical analysis of the data.

In [30]:

```
fig,ax = plt.subplots(1, 2)
titles = ['Twitter', 'YouTube']
dfs = [df_twitter, df_youtube]
for i in range(2):
    ax[i].hist(dfs[i]['polarity'], bins=20)
    ax[i].set_title(titles[i])
    ax[i].set_ylabel('Frequency')
    ax[i].set_xlabel('Polarity')
    ax[i].set_xlim((-1, 1))
plt.tight_layout()
```

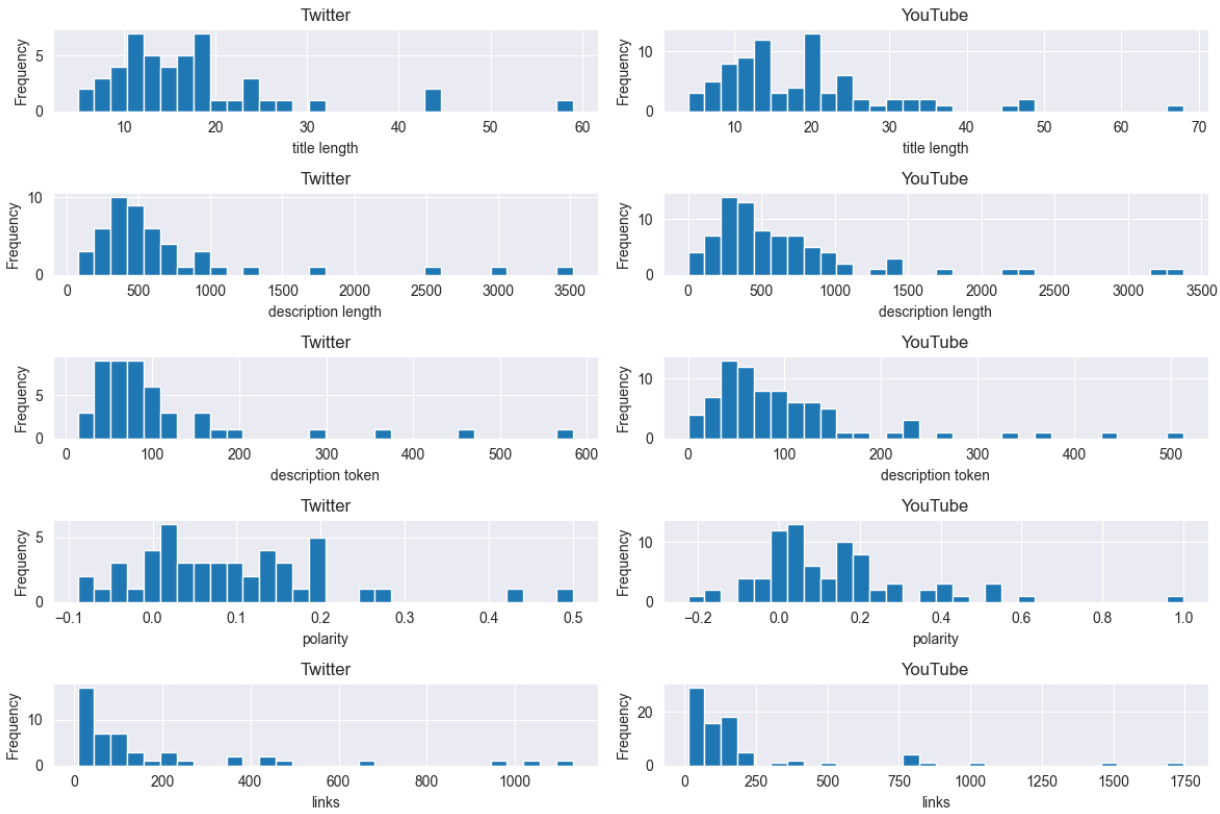


In [31]:
```
fig,ax = plt.subplots(1, 2)
titles = ['Twitter', 'YouTube']
dfs = [df_twitter, df_youtube]
for i in range(2):
    ax[i].hist(dfs[i]['links'], bins=50)
    ax[i].set_title(titles[i])
    ax[i].set_ylabel('Frequency')
    ax[i].set_xlabel('Links')
    ax[i].set_xlim((0, 2000))
plt.tight_layout()
```



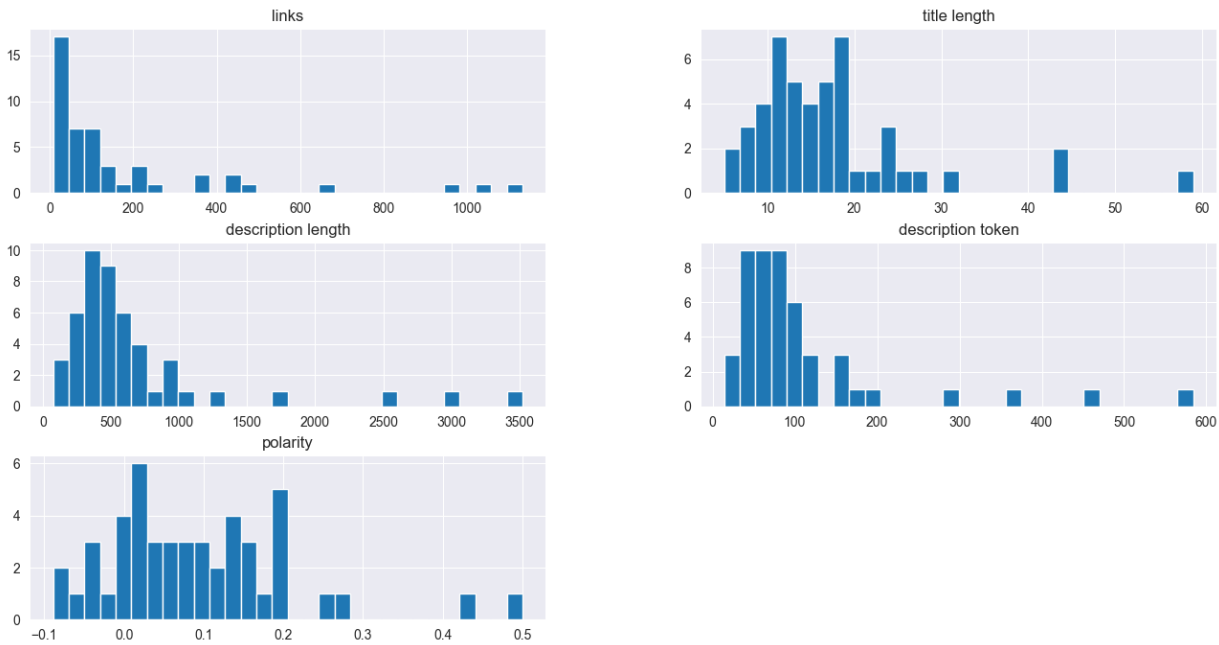
The following subplot provides an overview of the title length, description length, description token, polarity and links for both topics. On the left side the plots represent Twitter and on the right side YouTube. The data looks very similar to each other, a difference is the polarity with is higher for YouTube. This realisation was observed already before when looking at the statistical overview of the data.

In [86]:
```
features = ['title length', 'description length', 'description token', 'p
fig, ax = plt.subplots(len(features), 2)
titles = ['Twitter', 'YouTube']
dfs = [df_twitter, df_youtube]
for i in range(len(features)):
        for j in range(2):
                ax[i][j].hist(dfs[j][features[i]], bins=30)
                ax[i][j].set_title(titles[j])
                ax[i][j].set_ylabel('Frequency')
                ax[i][j].set_xlabel(features[i])
                # ax[i][j].set_xlim((-1, 1))
plt.tight_layout()
```



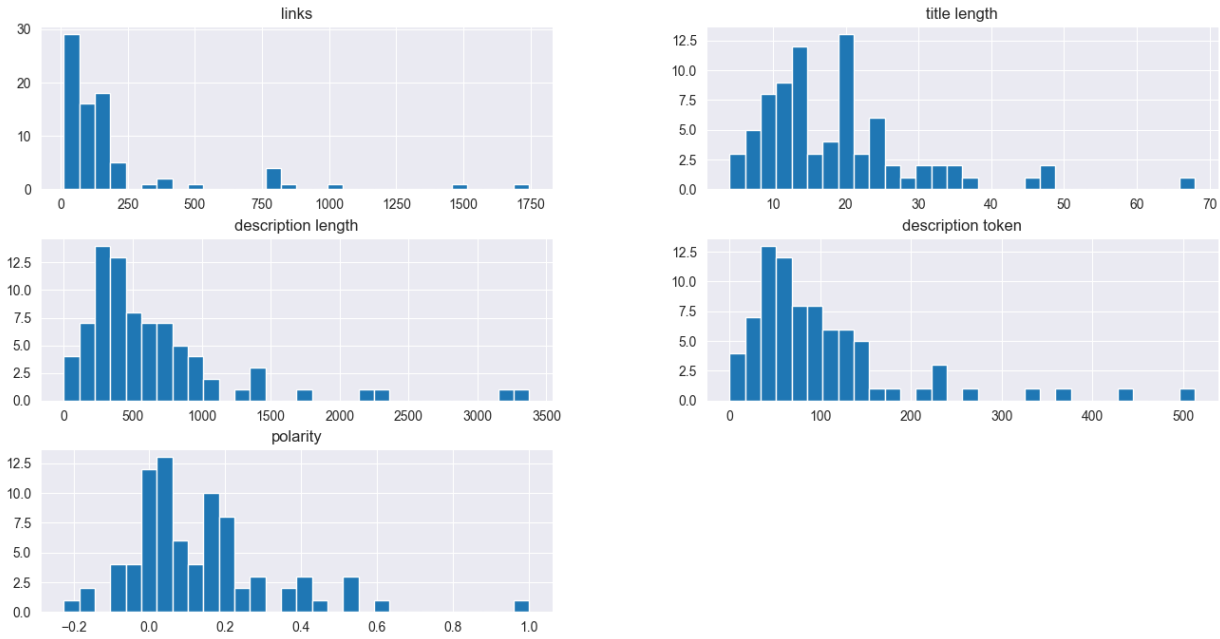
The histographs show again the same for Twitter.

In [89]:
```
df_twitter.hist(figsize=(16,8), bins=30);
```



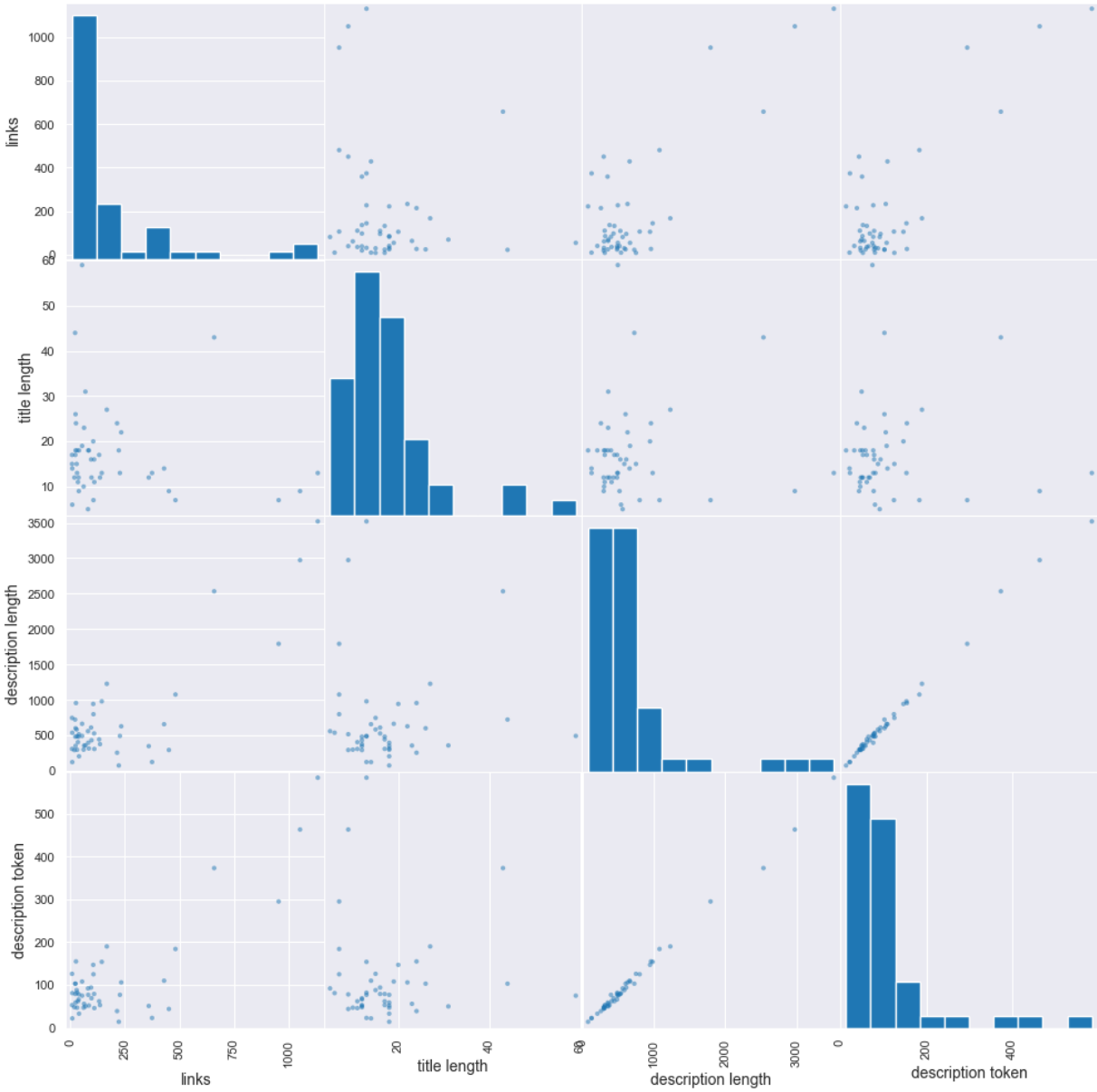
The histographs show again the same for YouTube.

In [90]:
```
df_youtube.hist(figsize=(16,8), bins=30);
```

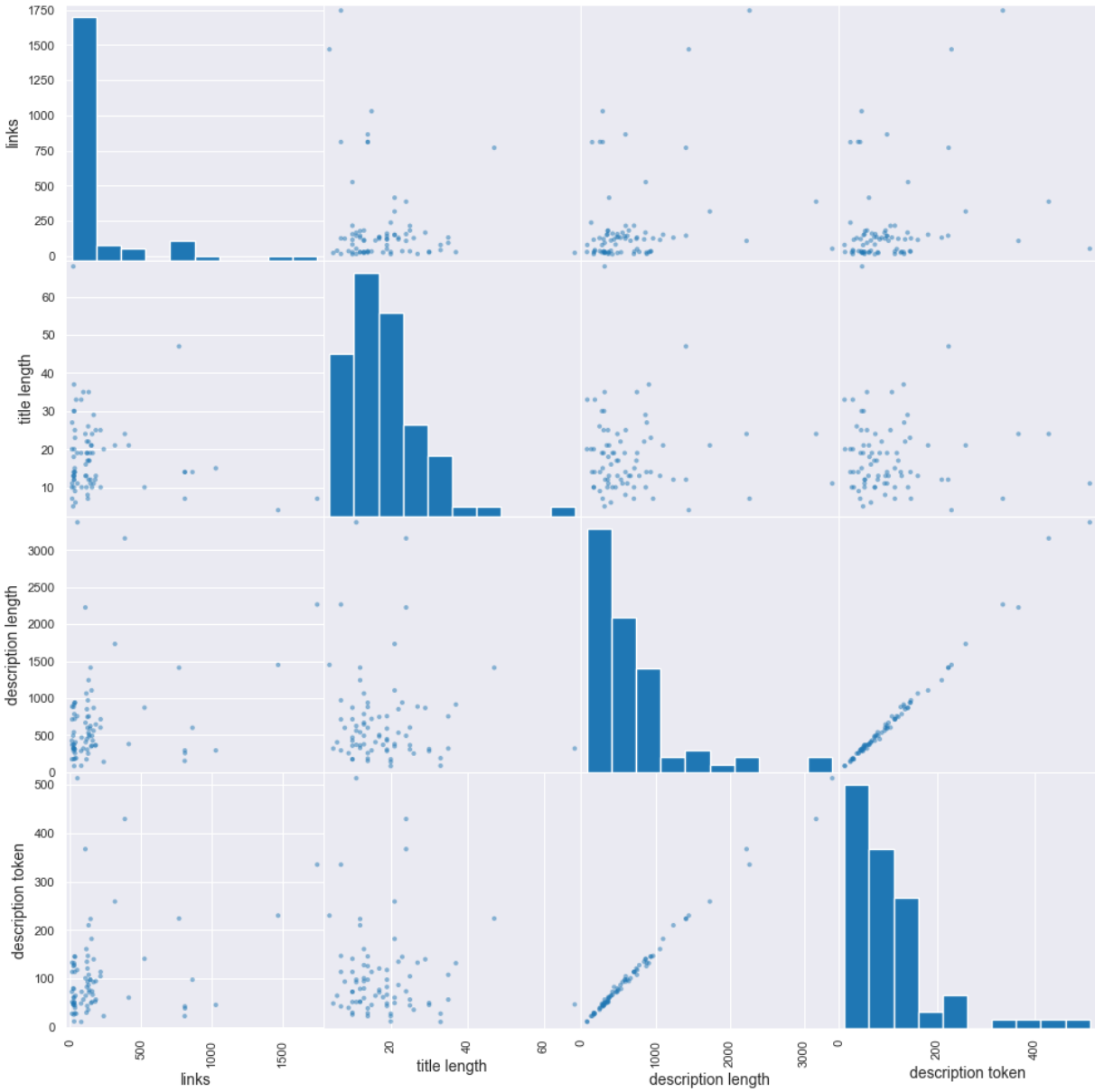


The scatter matrics displays the correlation between two features. A clear correlation between description length and description token can be seen. This was expectable, since both feature represent basically the same. For all other features no clear correlation can be observed.

In [35]:
```
pd.plotting.scatter_matrix(
    df_twitter.select_dtypes(include='number').iloc[:,-5:-1],
    figsize=(12,12)
);
```



In [36]:
```
pd.plotting.scatter_matrix(
    df_youtube.select_dtypes(include='number').iloc[:,-5:-1],
    figsize=(12,12)
);
```

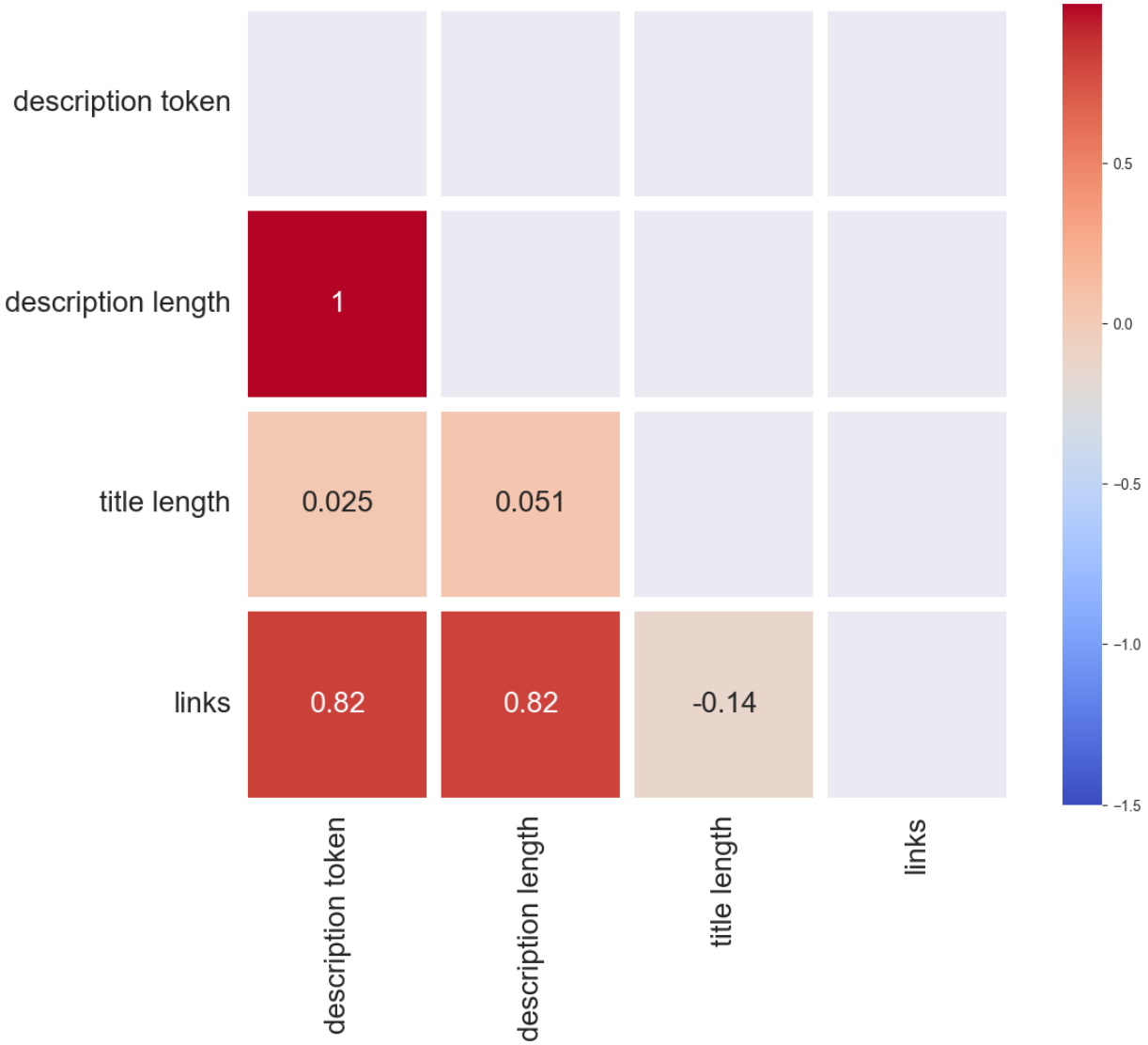


The correlation matrix shows for description length to description token a correlation of 1, which makes absolutely sense. It is not very surpricing that a correlation between links and description length/token can be found. Since it is likely that longer articles contains more links. However, this insight seems to hold true only for Twitter. YouTube has only a correlation of 0.26 between links and description length/token.

In [92]:
```
correlation = df_twitter[['description token', 'description length', 'tit
mask = np.zeros_like(correlation, dtype=np.bool)
mask[np.triu_indices_from(mask)] = True
plt.figure(figsize=(12,10))
plt.xticks(fontsize=20)
plt.yticks(fontsize=20)
sns.heatmap(correlation, cmap='coolwarm', annot=True, annot_kws={"size":
```

```
/var/folders/gf/mnrm20_51m9_99sdqvcn3bpw0000gn/T/ipykernel_8713/18036446
88.py:2: DeprecationWarning:

`np.bool` is a deprecated alias for the builtin `bool`. To silence this
warning, use `bool` by itself. Doing this will not modify any behavior a
nd is safe. If you specifically wanted the numpy scalar type, use `np.bo
ol_` here.
Deprecated in NumPy 1.20; for more details and guidance: https://numpy.o
rg/devdocs/release/1.20.0-notes.html#deprecations
```

Out[92]: <AxesSubplot: >



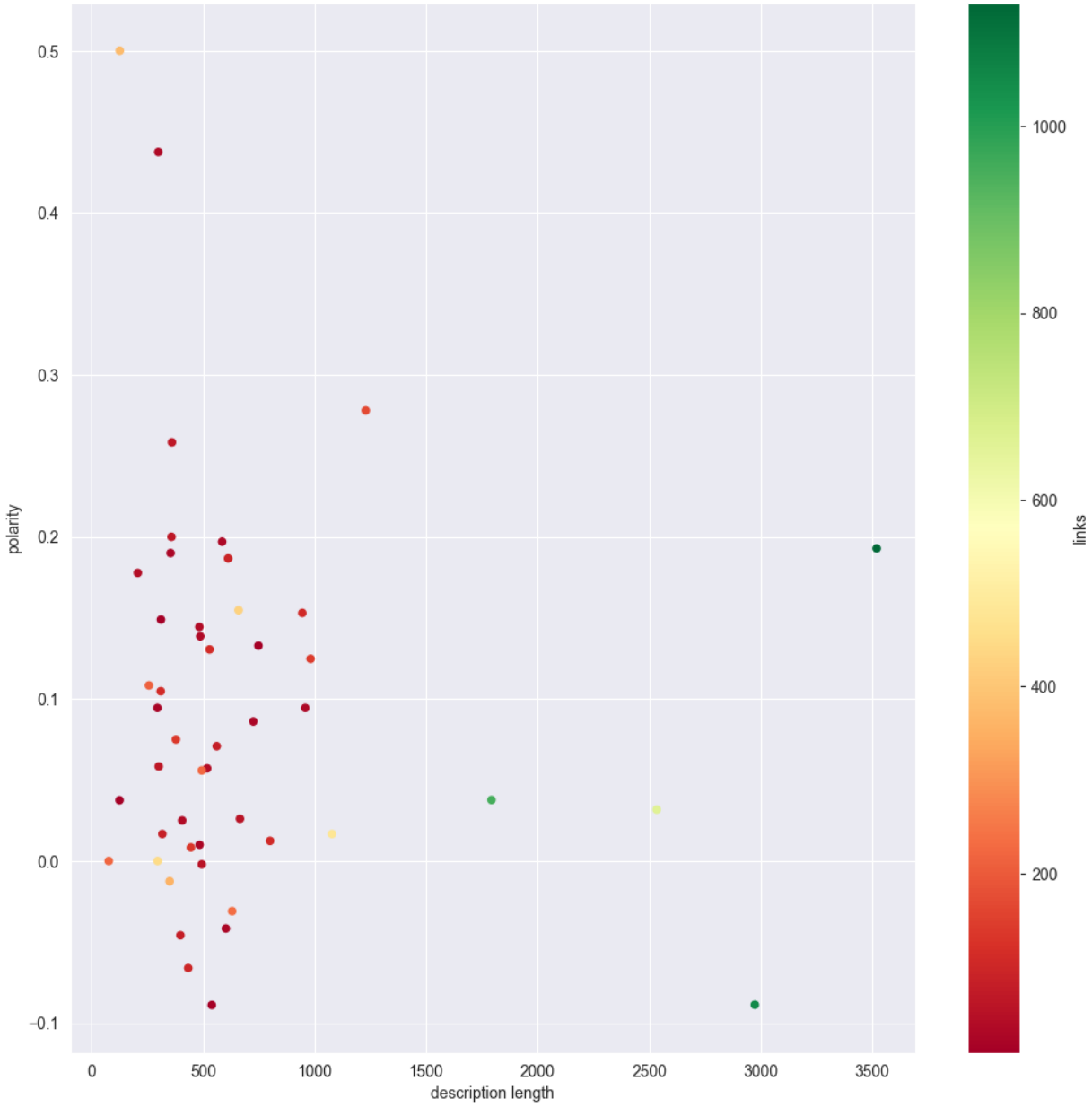
In [93]:
```
correlation = df_youtube[['description token', 'description length', 'tit
mask = np.zeros_like(correlation, dtype=np.bool)
mask[np.triu_indices_from(mask)] = True
plt.figure(figsize=(12,10))
plt.xticks(fontsize=20)
plt.yticks(fontsize=20)
sns.heatmap(correlation, cmap='coolwarm', annot=True, annot_kws={"size":
```

```
/var/folders/gf/mnrm20_51m9_99sdqvcn3bpw0000gn/T/ipykernel_8713/18279693
18.py:2: DeprecationWarning:

`np.bool` is a deprecated alias for the builtin `bool`. To silence this
warning, use `bool` by itself. Doing this will not modify any behavior a
nd is safe. If you specifically wanted the numpy scalar type, use `np.bo
ol_` here.
Deprecated in NumPy 1.20; for more details and guidance: https://numpy.o
rg/devdocs/release/1.20.0-notes.html#deprecations
```

Out[93]: <AxesSubplot: >

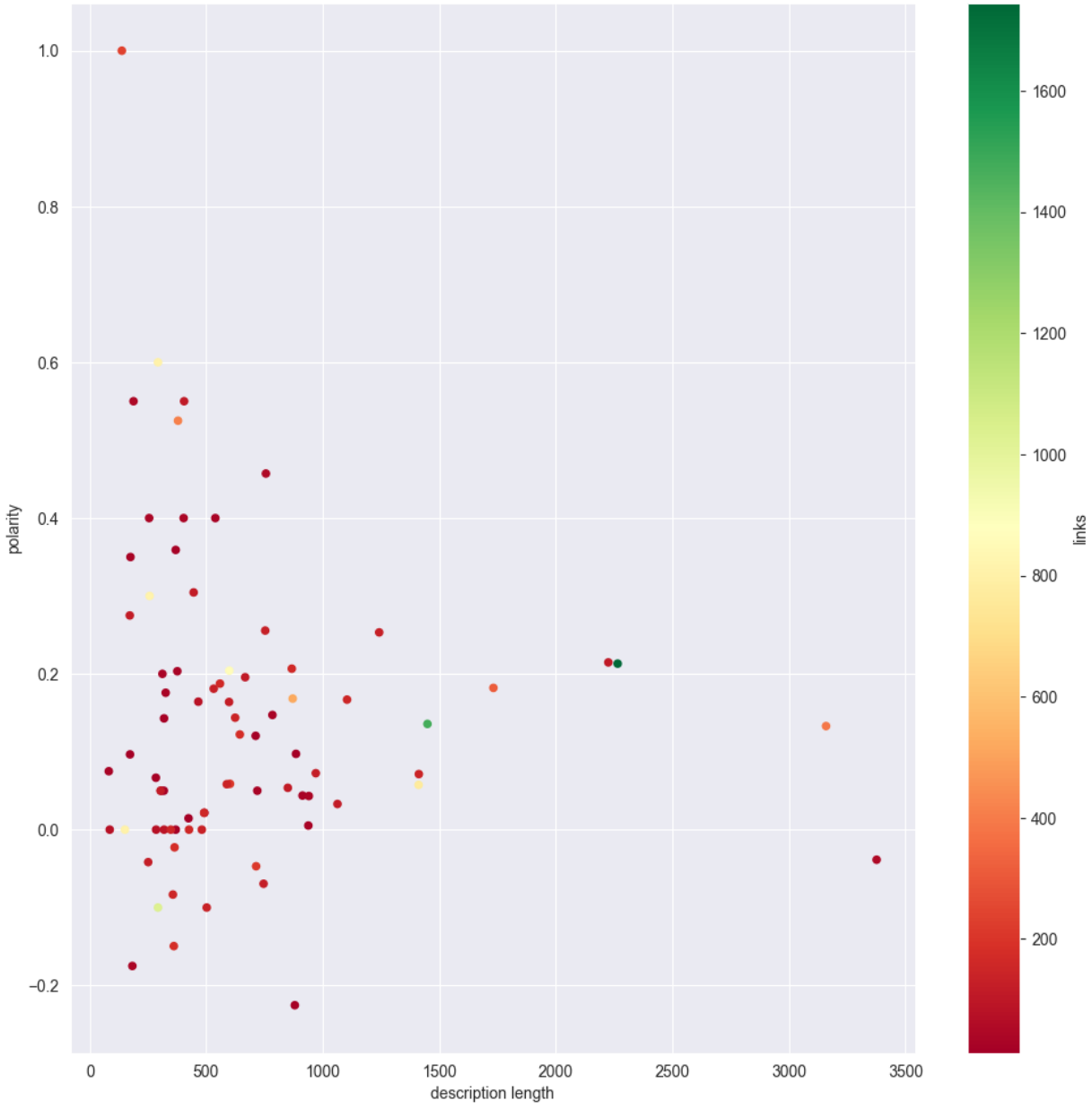
In [37]:
```
df_twitter.plot(kind='scatter', x='description length', y='polarity', c='
# plt.ylabel('Length')
# plt.ylabel('Polarity')
```

Out[37]: <AxesSubplot: xlabel='description length', ylabel='polarity'>



In [38]:
```
df_youtube.plot(kind='scatter', x='description length', y='polarity', c='
# plt.ylabel('Length')
# plt.ylabel('Polarity')
```

Out[38]:
```
<AxesSubplot: xlabel='description length', ylabel='polarity'>
```



N-gramming

Unigrams with stopwords

The unigrams below shows the count of the words for Twitter and YouTube. In these cases the stop words are not droped from the list. As it can be seen the word 'the' occurs the most. These unigrams have no value since only general words are listet (stopwords)

In [39]:
```
def get_top_n_words(corpus, n=None):
    vec = CountVectorizer().fit(corpus)
    bag_of_words = vec.transform(corpus)
    sum_words = bag_of_words.sum(axis=0)
    words_freq = [(word, sum_words[0, idx]) for word, idx in vec.vocabula
    words_freq =sorted(words_freq, key = lambda x: x[1], reverse=True)
    return words_freq[:n]
```

In [40]:
```
common_words = get_top_n_words(df_twitter['description'], 20)

unigrams_w_stops_twitter = pd.DataFrame(common_words, columns = ['word' ,
unigrams_w_stops_twitter
```

Out[40]:

	word	count
0	the	305
1	and	171
2	of	146
3	in	118
4	twitter	96
5	to	96
6	by	69
7	is	68
8	on	64
9	with	51
10	as	46
11	was	43
12	for	43
13	or	35
14	that	33
15	an	31
16	it	30
17	social	28
18	his	28
19	he	26

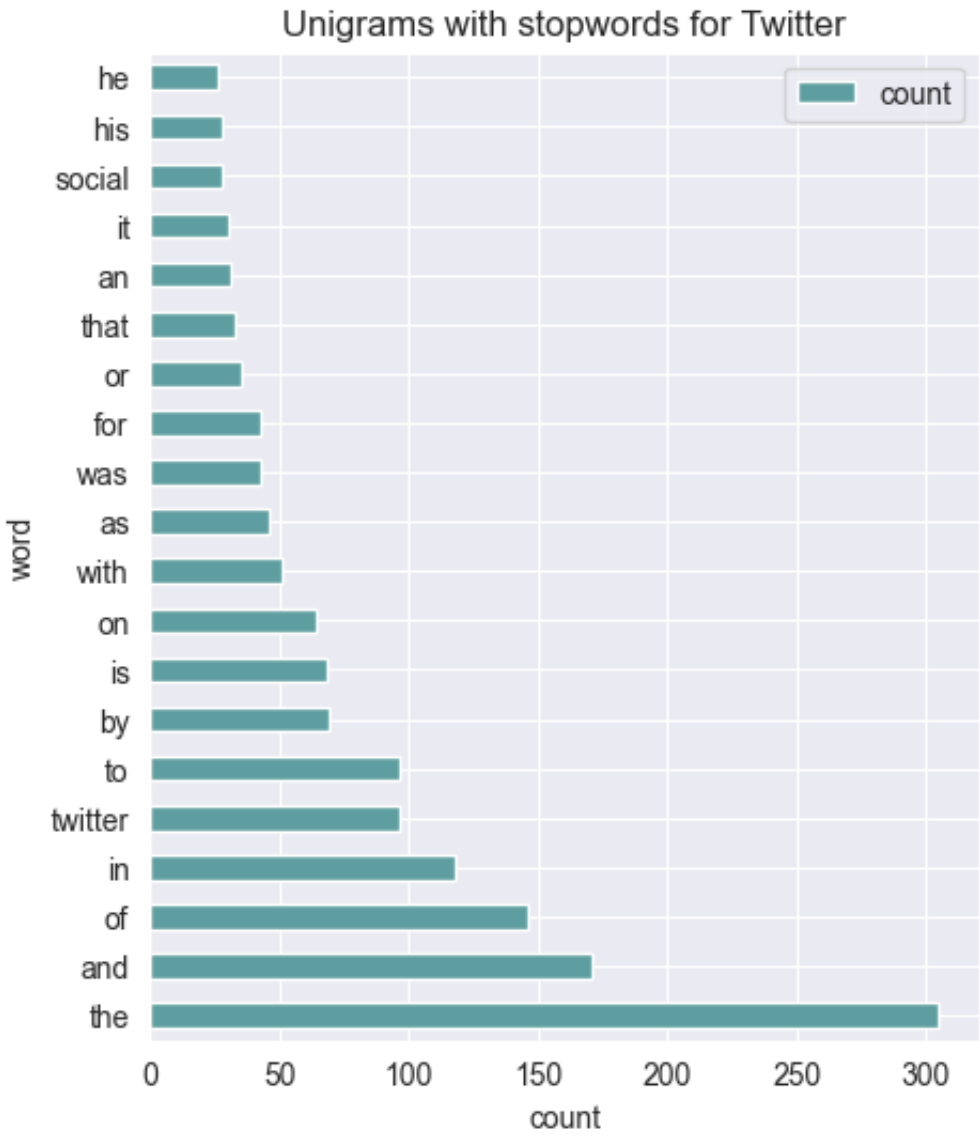
In [41]:
```
common_words = get_top_n_words(df_youtube['description'], 20)

unigrams_w_stops_youtube = pd.DataFrame(common_words, columns = ['word' ,
unigrams_w_stops_youtube
```

Out[41]:

	word	count
0	the	442
1	and	280
2	youtube	197
3	of	190
4	to	185
5	in	147
6	on	122
7	as	109
8	is	105
9	was	85
10	by	85
11	for	82
12	with	67
13	video	62
14	has	54
15	content	53
16	from	53
17	an	48
18	it	48
19	videos	43

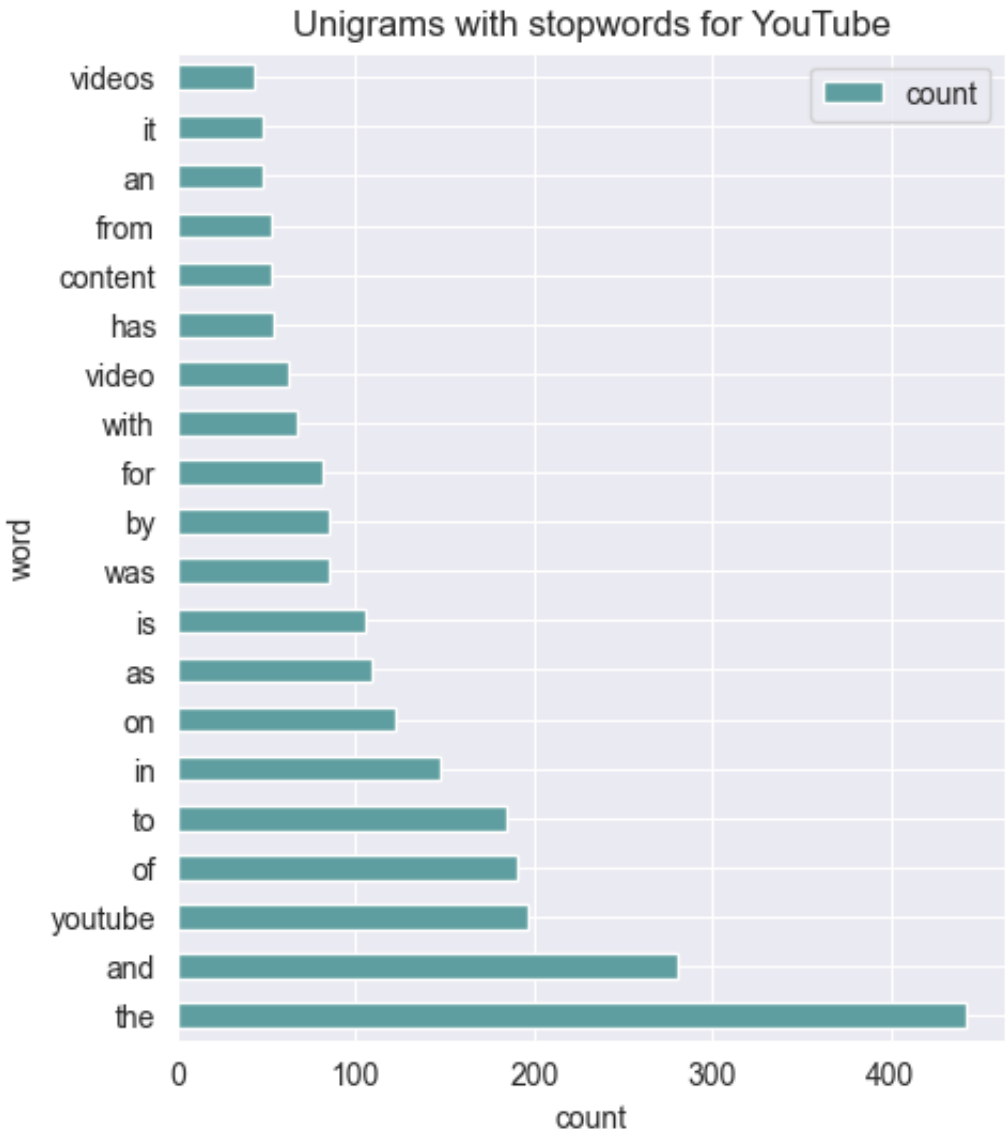
In [42]:

```
unigrams_w_stops_twitter.plot(kind='barh', x='word', color='cadetblue', w
plt.xlabel('count')
plt.ylabel('word')
plt.title('Unigrams with stopwords for Twitter')
plt.show()
```



In [43]:
```
unigrams_w_stops_youtube.plot(kind='barh', x='word', color='cadetblue', w
plt.xlabel('count')
plt.ylabel('word')
plt.title('Unigrams with stopwords for YouTube')
```

Out[43]:
```
Text(0.5, 1.0, 'Unigrams with stopwords for YouTube')
```



Unigrams without stopwords

In the example onwards the stopwords are droped. This results into more valuable insights of the data.

In [51]:
```
# Unigrams after stopword removal
def get_top_n_words(corpus, n=None):
    vec = CountVectorizer(stop_words = 'english').fit(corpus)
    bag_of_words = vec.transform(corpus)
    sum_words = bag_of_words.sum(axis=0)
    words_freq = [(word, sum_words[0, idx]) for word, idx in vec.vocabula
    words_freq =sorted(words_freq, key = lambda x: x[1], reverse=True)
    return words_freq[:n]
```

In [52]:
```
common_words = get_top_n_words(df_twitter['description'], 20)
unigrams_twitter = pd.DataFrame(common_words, columns = ['word' , 'count'
unigrams_twitter
```

Out[52]:

	word	count
0	twitter	96
1	social	28
2	tweets	24
3	users	18
4	media	17
5	world	15
6	series	15
7	musk	14
8	company	13
9	bieber	13
10	time	13
11	awards	13
12	100	12
13	hashtag	12
14	2015	11
15	protests	11
16	post	10
17	2009	10
18	trending	10
19	election	10

In [53]:

```
common_words = get_top_n_words(df_youtube['description'], 20)
unigrams_youtube = pd.DataFrame(common_words, columns = ['word' , 'count'
unigrams_youtube
```

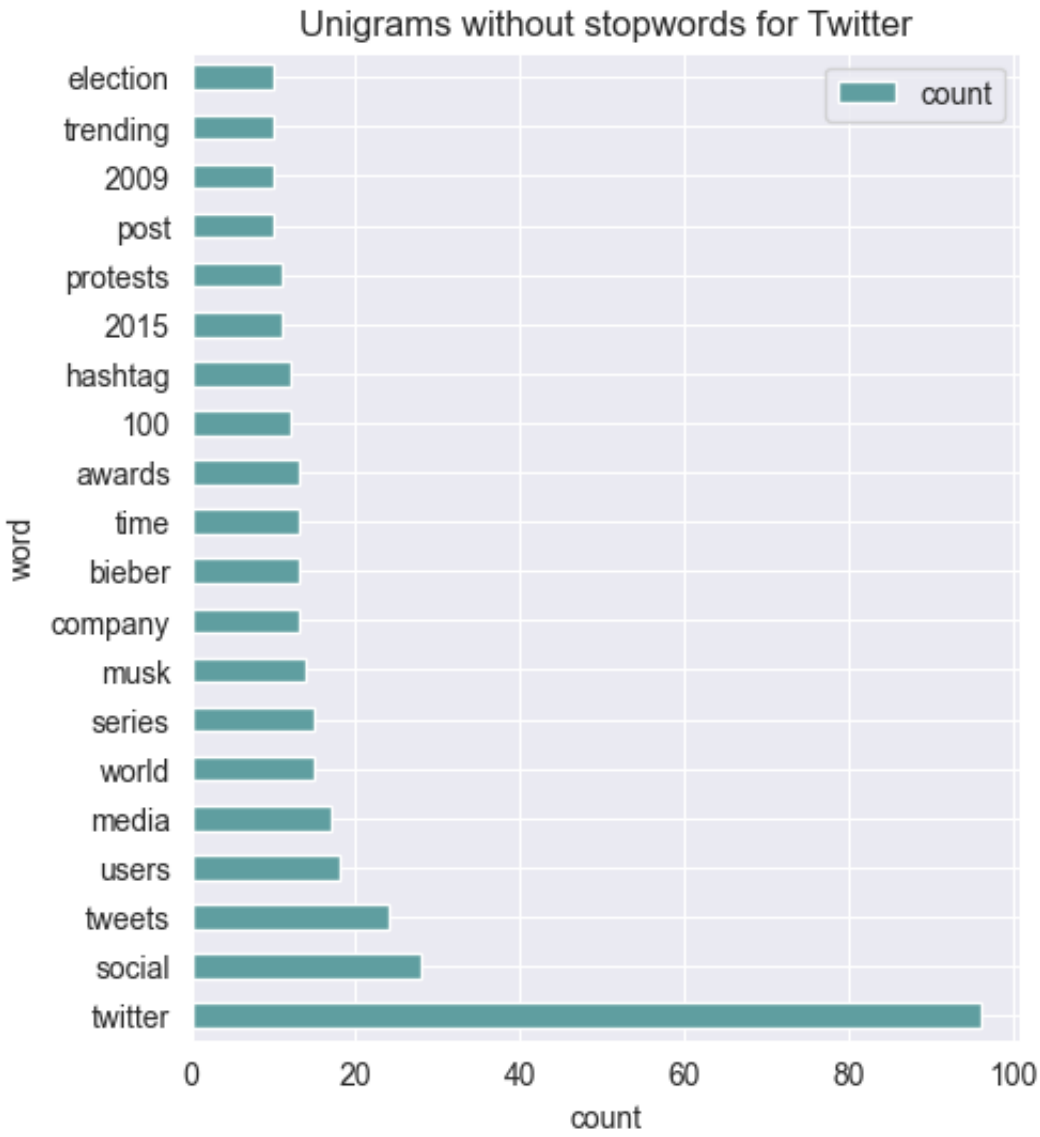
Out[53]:

	word	count
0	youtube	197
1	video	62
2	content	53
3	videos	43
4	music	36
5	channel	36
6	google	35
7	event	30
8	series	26
9	website	22
10	app	20
11	american	19
12	including	19
13	online	18
14	service	17
15	2019	16
16	lion	16
17	live	15
18	new	15
19	company	15

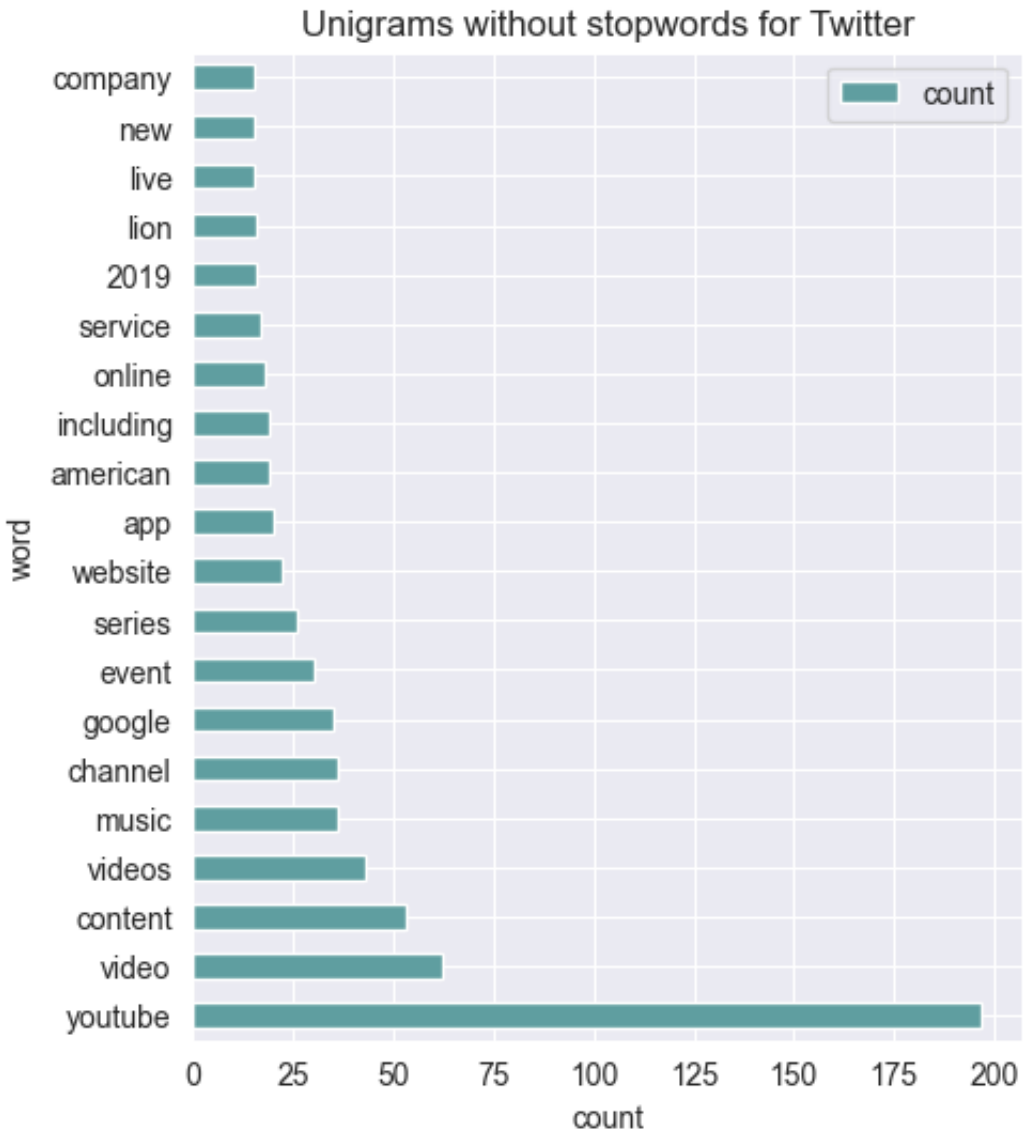
The unigrams show clearly which words are used the most. A very similar finding could already be found with the Word Cloud performed above.

In [54]:

```
unigrams_twitter.plot(kind='barh', x='word', color='cadetblue', width=0.5
plt.xlabel('count')
plt.ylabel('word')
plt.title('Unigrams without stopwords for Twitter')
plt.show()
```



In [55]:
```
unigrams_youtube.plot(kind='barh', x='word', color='cadetblue', width=0.5
plt.xlabel('count')
plt.ylabel('word')
plt.title('Unigrams without stopwords for Twitter')
plt.show()
```



Bigrams

The bigrams shows the same as the unigram, but in this case, two words are always taken together.

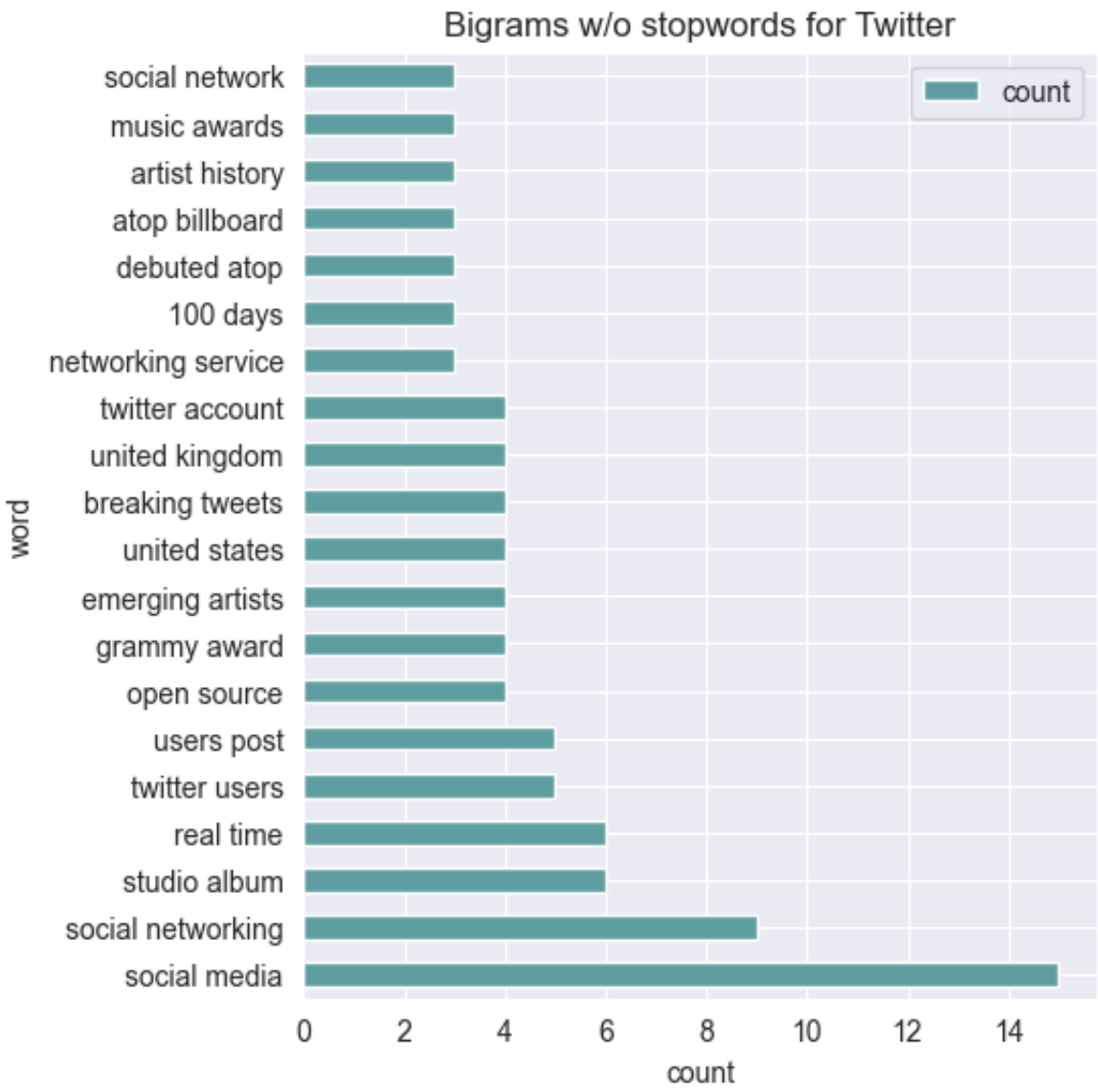
In [56]:
```
def get_top_n_bigram(corpus, n=None):
    vec = CountVectorizer(ngram_range=(2, 2), stop_words='english').fit(c
    bag_of_words = vec.transform(corpus)
    sum_words = bag_of_words.sum(axis=0)
    words_freq = [(word, sum_words[0, idx]) for word, idx in vec.vocabula
    words_freq =sorted(words_freq, key = lambda x: x[1], reverse=True)
    return words_freq[:n]
```

In [57]:
```
common_words = get_top_n_bigram(df_twitter['description'], 20)
bigrams_twitter = pd.DataFrame(common_words, columns = ['word' , 'count']
```

In [58]:
```
common_words = get_top_n_bigram(df_youtube['description'], 20)
bigrams_youtube = pd.DataFrame(common_words, columns = ['word' , 'count']
```

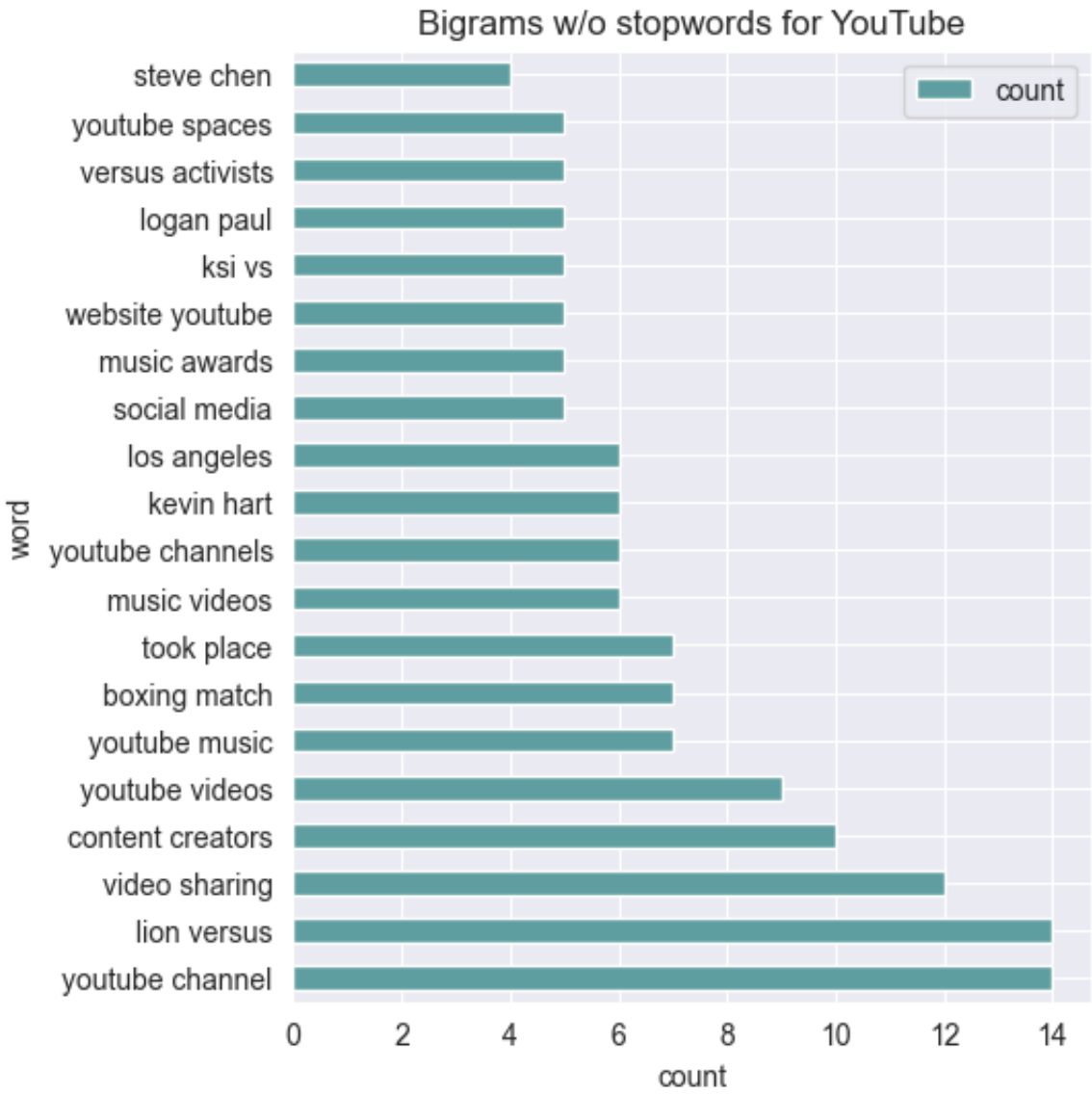
In [59]:
```
bigrams_twitter.plot(kind='barh', x='word', color='cadetblue', width=0.5,
plt.xlabel('count')
plt.ylabel('word')
plt.title('Bigrams w/o stopwords for Twitter')
```

Out[59]: Text(0.5, 1.0, 'Bigrams w/o stopwords for Twitter')



In [60]:
```
bigrams_youtube.plot(kind='barh', x='word', color='cadetblue', width=0.5,
plt.xlabel('count')
plt.ylabel('word')
plt.title('Bigrams w/o stopwords for YouTube')
```

Out[60]: Text(0.5, 1.0, 'Bigrams w/o stopwords for YouTube')



Trigrams

In this case three words are taken together. Most probably not enough data is available, therefore the following graphs do not have any meaning.

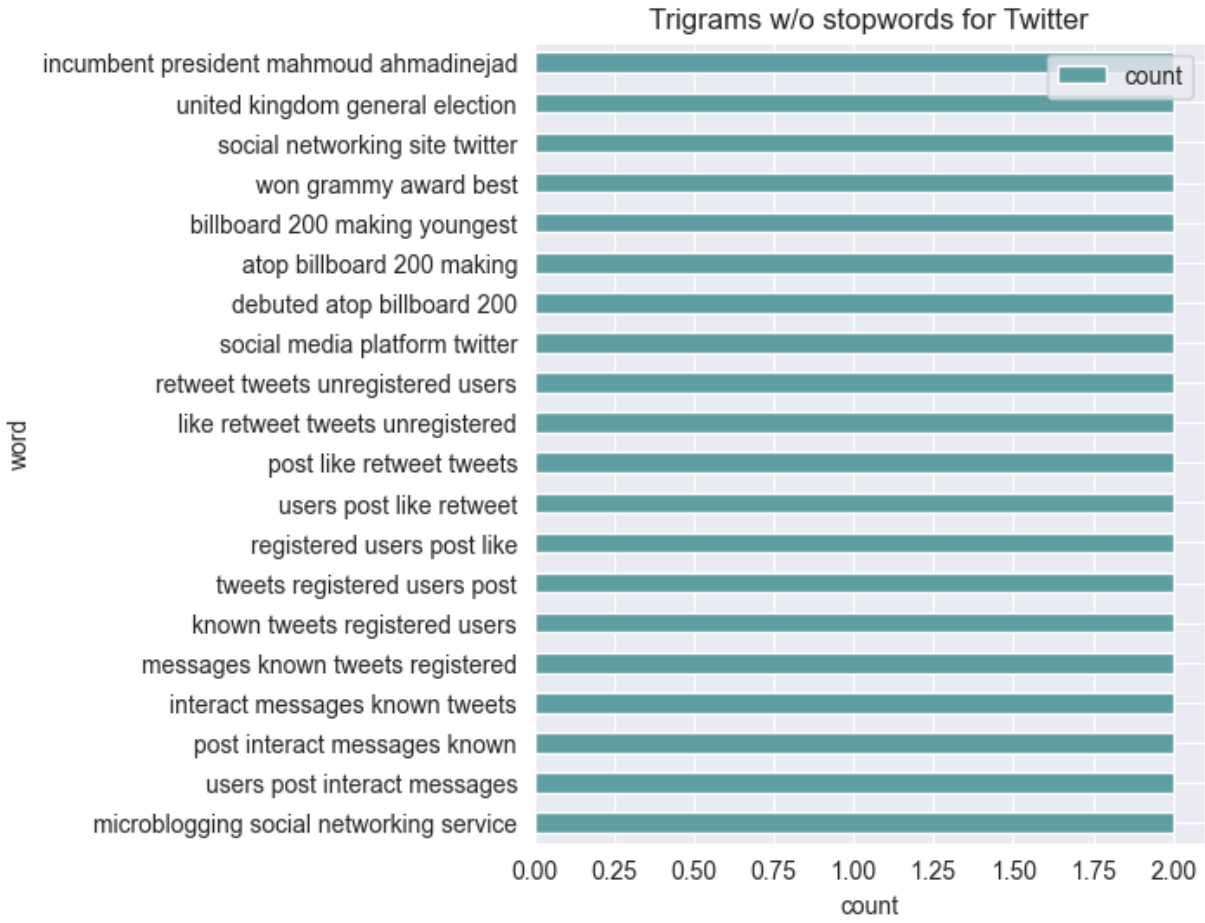
In [61]:
```
def get_top_n_trigram(corpus, n=None):
    vec = CountVectorizer(ngram_range=(4, 4), stop_words='english').fit(c
    bag_of_words = vec.transform(corpus)
    sum_words = bag_of_words.sum(axis=0)
    words_freq = [(word, sum_words[0, idx]) for word, idx in vec.vocabula
    words_freq =sorted(words_freq, key = lambda x: x[1], reverse=True)
    return words_freq[:n]
```

In [62]:
```
common_words = get_top_n_trigram(df_twitter['description'], 20)
trigrams_twitter = pd.DataFrame(common_words, columns = ['word' , 'count'
```

In [63]:
```
common_words = get_top_n_trigram(df_youtube['description'], 20)
trigrams_youtube = pd.DataFrame(common_words, columns = ['word' , 'count'
```

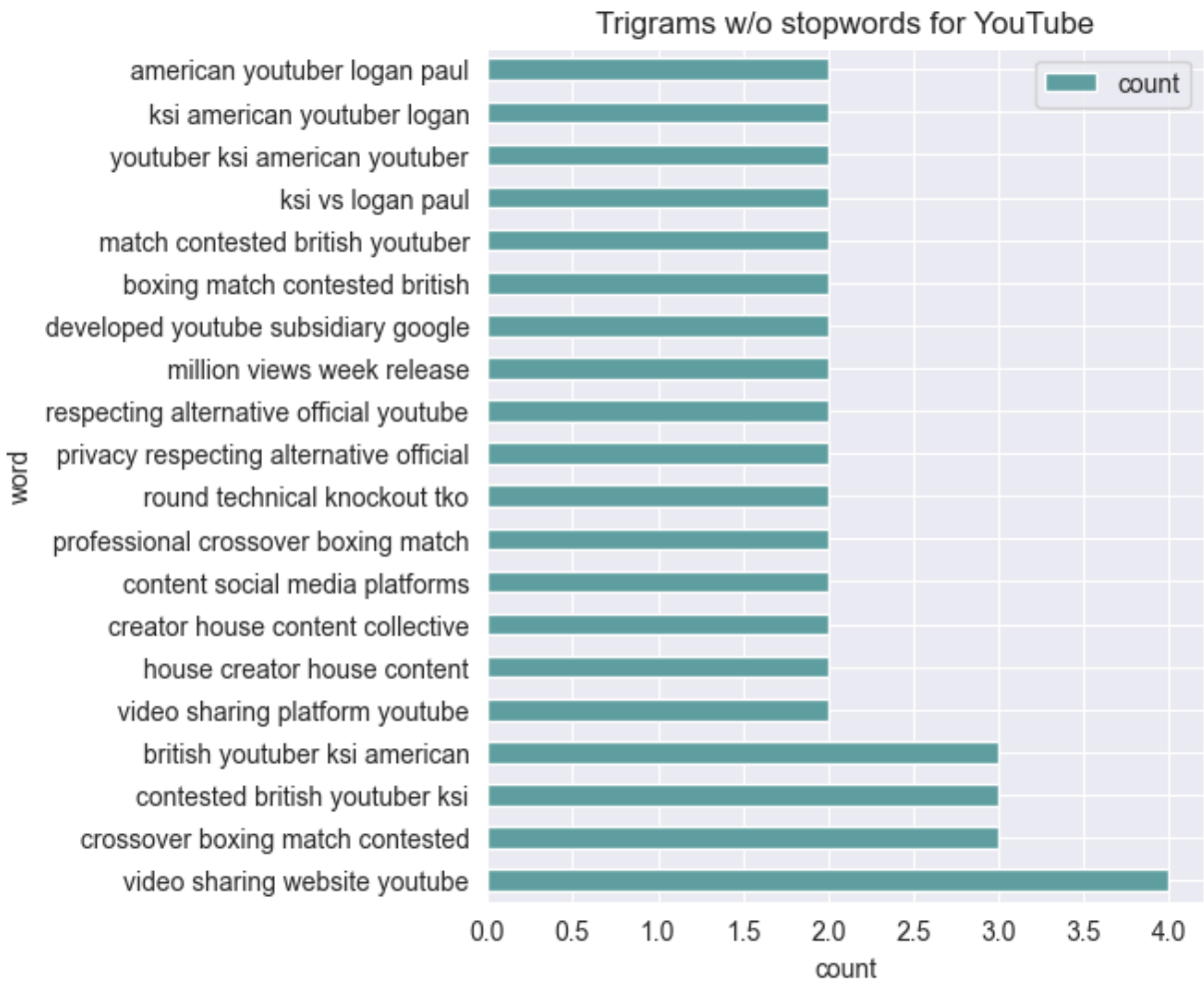
In [64]:
```
trigrams_twitter.plot(kind='barh', x='word', color='cadetblue', width=0.5
plt.xlabel('count')
plt.ylabel('word')
plt.title('Trigrams w/o stopwords for Twitter')
```

Out[64]:
```
Text(0.5, 1.0, 'Trigrams w/o stopwords for Twitter')
```



In [65]:
```
trigrams_youtube.plot(kind='barh', x='word', color='cadetblue', width=0.5
plt.xlabel('count')
plt.ylabel('word')
plt.title('Trigrams w/o stopwords for YouTube')
```

Out[65]:
```
Text(0.5, 1.0, 'Trigrams w/o stopwords for YouTube')
```


Trigrams w/o stopwords for YouTube
count
american youtuber logan paul
ksi american youtuber logan
youtuber ksi american youtuber
ksi vs logan paul
match contested british youtuber
boxing match contested british
developed youtube subsidiary google
million views week release
respecting alternative official youtube
privacy respecting alternative official
round technical knockout tko
professional crossover boxing match
content social media platforms
creator house content collective
house creator house content
video sharing platform youtube
british youtuber ksi american
contested british youtuber ksi
crossover boxing match contested
video sharing website youtube
word
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
count

Summary

The above exercise is clearly valuable, however to get really fluent in performing such analysis it has to be conducted on a regular basis. A few functions were really self-explanatory. Some others where more difficult and a look in the documentation was neccessary.

The polarity method is a great tool when used on the right dataset, but for this dataset it did not make any sense. The wikipedia articles are written from a very objective perspectiv without judgment if something is good or bad. Therefore, the polarity was very close to zero. It would be really interesting to analyse Twitter tweets or YouTube comments with the same method. (I am not sure if YouTube comment make sense, since YouTube deletes almost all negative comments.)

The Word Cloud is a nice tool to gain a overview, but it is more to play around and not to get a clear insight. To get a clear insight it is better to use a unigram or a bigram. In my case the dataset was to small to perform the trigram and get useful information out of it.